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in

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ABSTRACT

The exponential proliferation of internet-based communication has established electronic mail as the most critical and ubiquitous medium for professional and personal interaction worldwide. As of 2024, over 361 billion emails are exchanged globally per day, a figure projected to exceed 408 billion by 2027 (Statista, 2024). This scale of digital communication, however, has simultaneously created a fertile ground for exploitation by threat actors who leverage email as the predominant delivery mechanism for phishing attacks, ransomware campaigns, business email compromise (BEC), and Advanced Persistent Threats (APTs). IBM's Cost of a Data Breach Report (2023) confirms that phishing remains the leading initial attack vector, accounting for 16% of all enterprise breaches at an average remediation cost of USD 4.76 million per incident.

This report documents the research, conceptualization, design, implementation, and experimental evaluation of AetherGuard, an offline, multi-vector, heuristic-based Email Phishing Detector and Security Operations Center (SOC) simulation tool developed during an industry internship at Codec Technologies, Ahmedabad, spanning the period 19 January 2025 to 24 April 2025.

The central technical contribution of AetherGuard is a three-pillar Heuristic Threat Scoring Engine that analyses raw email payloads across: (1) Header Structural Integrity Analysis, which inspects RFC 5322 email metadata for mismatched sender domains, Authentication-Results failures (SPF, DKIM, DMARC), and anomalous relay routing; (2) Multi-Criteria URL Threat Analysis, which employs Python regular expressions to extract all embedded hyperlinks and evaluates each against eight evidence-based risk rules including IP-based URL masking, URL-shortener obfuscation, dangerous Top-Level Domain (TLD) patterns, and mismatched display-versus-destination links; and (3) Linguistic and Psychological Sentiment Profiling, which performs frequency-weighted scanning of the email body for urgency-weaponizing phrases, authority-spoofing language patterns, and coercive credential-harvesting triggers aligned with the MITRE ATT&CK framework techniques T1566.001 and T1566.002.

The scoring engine produces a normalised Composite Threat Index between 0 and 100, classifying the analysed email as Safe (0-30), Suspicious (31-70), or Critical/Phishing (71-100). The backend Python script is complemented by a fully self-contained, browser-based SOC dashboard GUI developed in HTML5, CSS3, and JavaScript, featuring a holographic-aesthetic interface with animated SVG threat Score Dial, Defensive Playbook generation, global

telemetry dashboard, and APT Intelligence Database.

Experimental validation across a curated dataset of 100 emails yielded an overall classification accuracy of 89%, a True Positive Rate (Recall) of 92%, a Precision rate of 86.8%, and an F1-Score of 89.3%. These results demonstrate the practical viability of heuristic-based detection as an accurate, privacy-preserving, zero-dependency, locally-deployable supplement or replacement to expensive cloud-based email security gateways in SME and academic environments.

Throughout the internship, comprehensive practical training was additionally undertaken in three supplementary cybersecurity disciplines: Network Traffic Analysis using Wireshark (including TCP analysis, DNS exfiltration detection, and HTTP credential capture); Incident Response Playbook development strictly aligned with the NIST Special Publication 800-61 Rev. 2 framework; and Digital Forensics Investigation utilizing FTK Imager, Autopsy, The Sleuth Kit, and the Volatility Framework to establish forensically sound post-breach investigation workflows.

Keywords: Phishing Detection, Email Security, Heuristic Analysis, Cybersecurity, Python, Flask, SOC Dashboard, MITRE ATT&CK, Network Traffic Analysis, Digital Forensics, Incident Response, NIST Framework, NLP, Threat Scoring.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
APT	Advanced Persistent Threat
CLI	Command-Line Interface
C2	Command and Control
DBIR	Data Breach Investigations Report
DFIR	Digital Forensics and Incident Response
DKIM	DomainKeys Identified Mail
DMARC	Domain-based Message Authentication Reporting and Conformance
DNS	Domain Name System
EDR	Endpoint Detection and Response
FPR	False Positive Rate
FNR	False Negative Rate
FTK	Forensic Toolkit
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IoC	Indicator of Compromise
IDS	Intrusion Detection System
IMAP	Internet Message Access Protocol
IP	Internet Protocol
IR	Incident Response
MITRE	MITRE Corporation (ATT&CK Framework)

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CHAPTER 1 - INTRODUCTION

1.1 Overview of Internship

This report documents the work undertaken during a three-month Cyber Security Internship at **Codec Technologies Pvt. Ltd.**, conducted in hybrid mode from **January 19, 2026 to April 24, 2026**. The internship was formally recognized under the All India Council for Technical Education (AICTE ID: CORPORATE6759d549ce59e1733940555) and accredited by the International Certification Accreditation Council (ICAC), under the mentorship of **Dr. Anurag Shrivastava, Program Manager, Codec Technologies Pvt. Ltd.**. The hybrid engagement model combined structured virtual expert-led learning sessions with hands-on applied project development in direct collaboration with the organization's in-house cybersecurity professionals.

The internship was organized in two distinct phases. The first phase spanning the initial month constituted a structured technical induction covering cybersecurity fundamentals, globally recognized security frameworks, defensive and offensive security operations, network security, threat intelligence, cloud security, and applied cryptography. This phase was designed to bring the intern to a professional baseline aligned with the competencies expected of an entry-level security analyst in the industry. The second phase spanning months two and three transitioned entirely into supervised project development, where the intern designed and delivered cybersecurity solutions addressing real-world threat scenarios, guided by Codec Technologies' security experts.

The declared domain of specialization for this internship is **Threat Detection and Defensive Security**, with a focused application in email-based threat analysis and phishing detection. The capstone deliverable of this engagement is **AetherGuard** an enterprise-grade Email Security Gateway with AI-powered phishing detection which forms the primary subject of this report.

1.2 About the Organization

Codec Technologies Pvt. Ltd. is a technology-driven organization specializing in professional cybersecurity skill development, applied training programs, and industry-aligned internship engagements for engineering students and IT professionals across India. Operating under formal AICTE recognition and ICAC accreditation, the organization bridges the gap between academic cybersecurity education and practical industry requirements by designing internship programs that simulate real enterprise security environments.

The internship curriculum at Codec Technologies is structured around globally recognized security frameworks and standards including the NIST Cybersecurity Framework, ISO/IEC 27001, OWASP Top 10, and GDPR compliance principles. Interns are mentored by practicing cybersecurity professionals throughout the project development phase, receiving guidance on system architecture, threat modeling, secure development practices, and professional deliverable validation — directly replicating the workflow of a functional security operations or product security team. This approach ensures that graduates of the program are equipped not just with theoretical knowledge, but with demonstrable project portfolios reflecting industry-grade standards.

1.3 Domain Background

Cybersecurity is one of the fastest-growing and most strategically critical disciplines in the modern digital economy. The proliferation of enterprise communication platforms, cloud-hosted services, remote work infrastructure, and interconnected systems has dramatically expanded the attack surface exploitable by threat actors. According to IBM Security's Cost of a Data Breach Report (2023), the global average cost of a data breach reached USD 4.45 million the highest ever recorded representing a 15% increase over three consecutive years.

Within the broader cybersecurity landscape, **defensive security** and **threat detection** represent the foundational pillars of enterprise protection strategy. Defensive security encompasses the systems, processes, and technologies deployed to monitor, detect, analyze, and respond to threats before they cause organizational damage. Key sub-disciplines include Security Information and Event Management (SIEM), endpoint detection and response (EDR), network traffic monitoring, email security, and incident response. Among these, email security has emerged as a particularly high-priority domain email remains the most widely abused vector for cyberattacks globally, with phishing responsible for initiating over 36% of all confirmed data breaches according to the Verizon Data Breach Investigations Report (DBIR, 2023).

Phishing attacks have grown significantly more sophisticated in recent years, leveraging AI-generated content, homograph domain spoofing, SPF/DKIM authentication bypass, and multi-redirect URL chains to evade conventional detection systems. This evolution has created a pressing need for intelligent, multi-layered email threat detection systems capable of combining machine learning classification with protocol-level authentication analysis and real-time URL reputation intelligence.

1.4 Problem Statement

Despite the widespread deployment of commercial email security gateways, phishing attacks continue to succeed at an alarming rate primarily because most available solutions rely on single-vector detection mechanisms such as keyword matching or signature-based filtering alone. Attackers exploit this limitation by crafting emails that pass individual security checks while remaining malicious when evaluated holistically. Furthermore, enterprise-grade email security platforms remain financially inaccessible to small and medium organizations, leaving a significant portion of the industry without adequate protection.

A critical technical gap exists in the availability of open, deployable, and explainable email threat detection tools that combine machine learning classification, email header authentication analysis (SPF, DKIM, DMARC), and real-time URL threat intelligence into a single unified platform. The absence of transparent, per-layer risk scoring in existing lightweight tools further limits the ability of security analysts to understand, validate, and act upon threat verdicts. AetherGuard was developed during this internship to directly address these gaps delivering a multi-layered, explainable, and fully offline-capable phishing detection gateway accessible to organizations of any size.

1.5 Scope of Work

The technical scope of this internship encompasses the following activities, conducted across the three-month engagement period:

- **Structured Security Training (Month 1):** Completed expert-delivered modules covering cybersecurity fundamentals, CIA Triad, NIST and ISO/IEC 27001 frameworks, networking security, SIEM operations, endpoint security, VAPT methodologies, OWASP Top 10 web application vulnerabilities, threat intelligence, incident response, cloud security, and cryptography
- **Project 1 — Network Traffic Analysis (Preliminary Exercise):** Captured and analyzed live network packets using Wireshark; examined DNS resolution, TCP handshake sequences, and TLS encrypted traffic behavior to develop practical network analysis skills
- **Project 2 — Secure Web Application Development (Supporting Project):** Developed an OWASP-aligned Flask web application; identified and mitigated SQL Injection and Broken Authentication vulnerabilities through parameterized queries, JWT authentication, and bcrypt-based password hashing

- **Project 3 — MobiSec Aegis Framework (Supporting Project):** Built a mobile application security testing dashboard integrating MobSF static analysis, Frida-based dynamic instrumentation, and ADB storage inspection — delivering automated vulnerability reporting via a Python Flask and React.js interface
- **Project 4 — AetherGuard: Email Security Gateway with Phishing Detection (Capstone):** Designed and implemented a full-stack, enterprise-grade phishing detection platform featuring a three-layer detection engine combining ML-based text classification (87.5% test accuracy), SPF/DKIM/DMARC header authentication parsing, and VirusTotal API-integrated URL reputation analysis. Supported multi-format scanning (.eml, .txt, .pdf, .docx) and delivered JSON-format incident reporting

The primary focus of this report is the design, implementation, and evaluation of **AetherGuard**, which represents the capstone technical contribution of this internship engagement.

CHAPTER 2 – AIMS AND OBJECTIVES



Figure 2.1 Internship Timeline

2.1 Aim

The primary aim of this internship was to develop professional-level competencies in cyber security through structured industry training and applied project development at Codec Technologies Pvt. Ltd.. Specifically, the internship sought to build practical expertise in the domain of **Threat Detection and Defensive Security** with a focused application in designing and implementing an intelligent email security system capable of detecting phishing threats using machine learning, email authentication protocols, and real-time threat intelligence. The engagement aimed to bridge the gap between academic knowledge and the operational demands of the cybersecurity industry by simulating the workflow, deliverables, and technical standards expected in a professional security environment.

2.2 Learning Objectives (Short-Term)

The following learning objectives guided the structured training phase of the internship, directly aligned with the curriculum delivered by Codec Technologies' cybersecurity professionals:

2.2.1 Cybersecurity Foundations and Frameworks

To develop a working understanding of the foundational principles of cybersecurity including the CIA Triad (Confidentiality, Integrity, Availability), common cyber threat categories and

attack vectors, and the operational principles of industry-recognized security frameworks including NIST Cybersecurity Framework, ISO/IEC 27001, and GDPR. This objective established the professional baseline from which all subsequent technical work was approached.

2.2.2 Network Security and Traffic Analysis

To acquire hands-on proficiency in network security fundamentals including TCP/IP protocol behavior, DNS resolution, HTTPS traffic inspection, and the configuration and evaluation of firewalls and Intrusion Detection/Prevention Systems (IDS/IPS). A specific sub-objective was to develop the ability to capture and interpret live network traffic using industry-standard tools such as Wireshark, enabling identification of protocol anomalies and suspicious network behavior.

2.2.3 Defensive Security Operations

To understand and apply the principles of enterprise defensive security including the setup and use of Security Information and Event Management (SIEM) platforms for real-time log aggregation, security incident detection, and response coordination. This objective also encompassed endpoint security practices including the deployment and evaluation of antivirus solutions, Endpoint Detection and Response (EDR) tools, and best-practice endpoint hardening techniques.

2.2.4 Vulnerability Assessment and Penetration Testing (VAPT)

To develop competency in systematic vulnerability assessment and penetration testing methodologies using industry-grade tools including Nessus, OpenVAS, Nikto, Metasploit, and Burp Suite. This objective included understanding the ethical and legal framework governing penetration testing, the structured execution of a VAPT engagement lifecycle from reconnaissance and scanning to exploitation and reporting and the interpretation of vulnerability severity using CVSS scoring.

2.2.5 Web Application Security

To apply OWASP Top 10 principles in identifying, exploiting, and remediating critical web application vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and Broken Authentication. A key practical objective was to validate security mitigations through controlled attack simulation, demonstrating both the vulnerability and its elimination in a working application.

2.2.6 Threat Intelligence and Incident Response

To understand the lifecycle of a cybersecurity incident from initial threat detection through containment, eradication, and post-incident analysis. This included studying structured threat intelligence collection and analysis practices, and understanding the role of threat feeds (such as VirusTotal) in enriching real-time detection systems directly applied in the AetherGuard capstone project.

2.2.7 Cloud Security and Cryptography

To gain foundational knowledge of cloud security architecture across major platforms (AWS, Azure, GCP), including the shared responsibility model, cloud-native security tooling, and common cloud misconfiguration risks. Additionally, to understand applied cryptography including symmetric and asymmetric encryption algorithms, hashing, SSL/TLS protocol operation, and VPN architectures providing the theoretical basis for evaluating and implementing secure communication in project work.

2.3 Project Objectives

The following technical objectives governed the project development phase of the internship:

- To design a multi-layered phishing detection architecture that combines machine learning classification, email header authentication analysis, and URL reputation intelligence into a single, unified threat assessment pipeline
- To train and validate a Logistic Regression classifier using TF-IDF bigram vectorization on a curated dataset of real-world phishing and legitimate email samples, achieving measurable detection accuracy
- To implement RFC 5322-compliant email parsing capable of extracting and evaluating SPF, DKIM, and DMARC authentication headers from raw .eml files
- To integrate VirusTotal API v3 for real-time URL reputation scoring, complemented by a local heuristic engine capable of detecting IP-based URLs, homograph domains, suspicious TLDs, URL shorteners, and redirect chains
- To develop a transparent, per-layer risk scoring system providing analysts with full explainability of every threat verdict generated
- To deliver a production-ready web interface with session telemetry, scan history persistence, and JSON-format incident report export replicating the output expected from an enterprise security operations tool

2.4 Long-Term Learning Outcomes

Beyond the immediate technical deliverables, this internship was designed to cultivate the following long-term professional outcomes:

- **Industry Readiness:** Develop the ability to independently scope, design, and deliver a security tool from requirement analysis through implementation and testing aligned with the workflow of a professional security engineer
- **Framework Literacy:** Establish working familiarity with NIST, ISO/IEC 27001, OWASP, and MITRE ATT&CK frameworks as operational reference standards rather than abstract theoretical constructs
- **Research Integration:** Apply academic knowledge of machine learning and security research to produce a working system with measurable, reproducible performance metrics bridging the gap between research and deployment
- **Security Mindset:** Internalize the defensive security mindset the habit of evaluating systems not only for functionality but for the attack surfaces they expose and the controls required to mitigate them
- **Professional Documentation:** Develop the ability to produce technical security reports, incident documentation, and system architecture descriptions meeting industry communication standards

CHAPTER 3 - REVIEW OF LITERATURE

3.1 Introduction to Cybersecurity

Cybersecurity is defined as the practice of protecting systems, networks, applications, and data from digital attacks, unauthorized access, damage, and disruption. As organizations increasingly depend on digital infrastructure for their core operations, the importance of cybersecurity has grown from a technical concern to a fundamental business and national security imperative. The discipline encompasses a wide range of sub-domains including network security, application security, endpoint protection, identity management, cryptography, threat intelligence, and incident response each addressing a distinct layer of the modern digital attack surface.

The foundational principle underpinning all cybersecurity practice is the **CIA Triad** a model that defines three core security objectives: Confidentiality, Integrity, and Availability. Confidentiality ensures that information is accessible only to those authorized to access it. Integrity guarantees that data remains accurate and unaltered except through authorized processes. Availability ensures that systems and data are accessible to authorized users when needed. All security controls, policies, and technologies are ultimately designed to protect one or more of these three properties (NIST, 2018).



Figure 3.1 — The CIA Triad: Core Principles of Cybersecurity

The global cybersecurity threat landscape has intensified considerably over the past decade. According to IBM Security's Cost of a Data Breach Report (2023), the average global cost of a data breach reached USD 4.45 million a 15% increase over three years and the highest figure ever recorded. The same report found that breaches involving stolen credentials, phishing, and cloud misconfiguration accounted for the majority of incidents by root cause. These statistics underscore the urgency of developing robust, multi-layered security architectures that address threats across all attack vectors.

3.2 Cybersecurity Frameworks and Standards

Standardized cybersecurity frameworks provide organizations with structured, repeatable approaches to managing cybersecurity risk. Three frameworks are most widely referenced in industry practice: the NIST Cybersecurity Framework, ISO/IEC 27001, and the General Data Protection Regulation (GDPR).

3.2.1 NIST Cybersecurity Framework (CSF)

The National Institute of Standards and Technology (NIST) Cybersecurity Framework, first published in 2014 and updated in 2018, provides a policy framework of computer security guidance for how private sector organizations can assess and improve their ability to prevent, detect, and respond to cyberattacks (NIST, 2018). The framework is organized around five core functions:

- **Identify** — Develop an organizational understanding of cybersecurity risk to systems, assets, and capabilities
- **Protect** — Implement appropriate safeguards to ensure delivery of critical infrastructure services
- **Detect** — Develop and implement appropriate activities to identify the occurrence of a cybersecurity event
- **Respond** — Develop and implement appropriate activities to take action regarding a detected cybersecurity incident
- **Recover** — Develop and implement appropriate activities to maintain plans for resilience and restore any capabilities impaired during a cybersecurity incident



Figure 3.2 — NIST CSF Core Functions

The NIST framework is technology-neutral and sector-agnostic, making it applicable to organizations of any size or industry. Its adoption within the internship curriculum at Codec Technologies ensures that project work including AetherGuard is evaluated against internationally recognized security management principles (Codec Technologies, 2026).

3.2.2 ISO/IEC 27001

ISO/IEC 27001 is an international standard for Information Security Management Systems (ISMS). It specifies the requirements for establishing, implementing, maintaining, and continually improving an information security management system within the context of an organization's risk management framework (ISO, 2022). Unlike NIST CSF, which is a framework, ISO/IEC 27001 is a certifiable standard organization can undergo formal audits and receive certification demonstrating compliance with its requirements.

Key control domains in ISO/IEC 27001 include information security policies, organization of information security, human resource security, asset management, access control, cryptography, physical and environmental security, operations security, communications security, and incident management. The standard's emphasis on documented risk assessment and treatment processes aligns closely with the structured threat modeling and security analysis practices employed during this internship.

3.2.3 General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR), enforced since May 2018, is a comprehensive European Union data privacy law that governs the collection, processing, storage, and transfer

of personal data of EU residents (European Parliament, 2016). From a cybersecurity perspective, GDPR mandates that organizations implement appropriate technical and organizational measures to ensure a level of security appropriate to the risk including protection against unauthorized access, accidental loss, and data breaches. Organizations must report data breaches to supervisory authorities within 72 hours of discovery, creating a direct operational requirement for the kind of threat detection and incident response capabilities developed in this internship.

3.3 Network Security and Traffic Analysis

Network security encompasses the policies, practices, and technologies deployed to monitor, detect, prevent, and respond to unauthorized access and attacks targeting network infrastructure. At its core, network security relies on an understanding of the fundamental protocols governing data communication particularly the TCP/IP model, DNS, and HTTP/HTTPS.

3.3.1 TCP/IP Protocol Suite

The TCP/IP protocol suite forms the backbone of modern internet communication. The Transmission Control Protocol (TCP) governs reliable, connection-oriented data transfer through a three-way handshake process (SYN → SYN-ACK → ACK) that establishes a verified communication session before data exchange begins. The Internet Protocol (IP) handles logical addressing and packet routing across heterogeneous networks. Understanding TCP/IP behavior is fundamental to network security analysis anomalies in handshake sequences, unusual port activity, and unexpected packet volumes are among the earliest indicators of network-level attacks including port scanning, SYN flooding, and man-in-the-middle attacks.

3.3.2 DNS and Its Security Implications

The Domain Name System (DNS) translates human-readable domain names into machine-readable IP addresses, functioning as the internet's directory service. From a security perspective, DNS is a frequent target and vector for attacks. DNS spoofing (cache poisoning) allows attackers to redirect users from legitimate domains to malicious servers. DNS tunneling exploits DNS query/response traffic to exfiltrate data or establish covert command-and-control channels bypassing conventional firewall rules. During the Network Traffic Analysis project conducted in this internship, DNS resolution patterns were captured and analyzed using Wireshark to identify normal versus anomalous domain lookup behavior.

3.3.3 TLS and Encrypted Traffic Inspection

Transport Layer Security (TLS) is the cryptographic protocol that provides confidentiality and integrity for data transmitted over the internet, replacing the deprecated SSL protocol. TLS operates through a certificate-based handshake that establishes an encrypted channel between client and server. While TLS protects the content of communications from interception, it also introduces a challenge for security monitoring encrypted traffic cannot be directly inspected without decryption, limiting the visibility of network-based security tools. Modern enterprise security architectures address this through TLS inspection proxies that decrypt, inspect, and re-encrypt traffic at the network perimeter. Traffic analysis conducted during this internship confirmed that the majority of observed network communications operated over HTTPS (port 443), consistent with modern web security standards.

3.4 Defensive Security Operations

Defensive security encompasses the technologies and processes deployed to monitor, detect, and respond to threats targeting an organization's infrastructure. Two foundational components of enterprise defensive security are Security Information and Event Management (SIEM) systems and Endpoint Detection and Response (EDR) solutions.

3.4.1 Security Information and Event Management (SIEM)

SIEM platforms aggregate, normalize, correlate, and analyze security log data from across an organization's entire technology stack including network devices, servers, applications, and security controls in real time. By correlating events across multiple sources, SIEM systems can identify attack patterns that would be invisible when examining any single log source in isolation. Modern SIEM platforms such as Splunk, IBM QRadar, and Microsoft Sentinel incorporate behavioral analytics and machine learning to detect anomalous activity indicative of insider threats, lateral movement, and advanced persistent threats (APTs). The study of SIEM operations during this internship provided foundational knowledge of how security analysts triage alerts, investigate incidents, and escalate confirmed threats through formal response workflows.

3.4.2 Endpoint Security and EDR

Endpoint Detection and Response (EDR) solutions provide continuous monitoring and response capabilities at the device level covering laptops, desktops, servers, and mobile devices. Unlike traditional antivirus solutions that rely on static signature databases, EDR tools use behavioral analysis to detect threats based on what processes are doing, rather than what

they look like. Key capabilities include process monitoring, file system activity tracking, network connection logging, and automated containment of suspected malicious activity. Leading EDR platforms include CrowdStrike Falcon, Microsoft Defender for Endpoint, and SentinelOne. The integration of endpoint visibility with SIEM telemetry forms the foundation of modern Security Operations Center (SOC) monitoring architectures.

3.5 Vulnerability Assessment and Penetration Testing (VAPT)

Vulnerability Assessment and Penetration Testing (VAPT) is a structured methodology for identifying, quantifying, and prioritizing security vulnerabilities in systems, networks, and applications. VAPT is a critical component of proactive security programs organizations that perform regular VAPT engagements identify and remediate vulnerabilities before attackers can exploit them.

3.5.1 Vulnerability Assessment

Vulnerability assessment involves the systematic scanning and enumeration of a target system to identify known security weaknesses. Tools commonly used in professional vulnerability assessments include Nessus, OpenVAS, and Nikto. These tools query systems against databases of known Common Vulnerabilities and Exposures (CVEs), identifying unpatched software, misconfigured services, and insecure default settings. Vulnerability findings are typically scored using the Common Vulnerability Scoring System (CVSS), a standardized framework that rates vulnerabilities on a 0–10 scale across metrics including attack complexity, required privileges, and potential impact.

3.5.2 Penetration Testing

Penetration testing goes beyond vulnerability assessment by actively attempting to exploit identified weaknesses simulating the actions of a real attacker within a controlled, authorized engagement. A standard penetration testing engagement follows five phases:

1. **Reconnaissance** — Gathering information about the target through passive (OSINT) and active techniques
2. **Scanning and Enumeration** — Identifying open ports, running services, and system configurations using tools like Nmap
3. **Exploitation** — Attempting to exploit identified vulnerabilities using frameworks such as Metasploit

4. **Post-Exploitation** — Assessing the extent of access achieved and potential for lateral movement
5. **Reporting** — Documenting all findings with severity ratings, proof-of-concept evidence, and remediation recommendations

The VAPT methodology studied during this internship provides the analytical foundation for evaluating security systems including the threat modeling approach applied in the design of AetherGuard's detection architecture.

3.6 Web Application Security and OWASP Top 10

Web application security addresses the vulnerabilities and attack techniques targeting web-based systems the most widely deployed and frequently targeted class of applications in the modern enterprise. The Open Web Application Security Project (OWASP) maintains the OWASP Top 10 a regularly updated consensus document representing the ten most critical web application security risks, widely recognized as the industry-standard reference for web security risk management (OWASP, 2021).



Figure 3.3 — OWASP Top 10 (2021): Most Critical Web Application Security Risks

In the Secure Web Application project, vulnerabilities A01 (Broken Access Control), A02 (Cryptographic Failures), and A03 (Injection) were intentionally implemented and subsequently mitigated through JWT-based authentication, script password hashing, and

parameterized queries using Flask-SQLAlchemy ORM — validating each mitigation through controlled attack simulation.

The ten categories most relevant to the Secure Web Application project developed during this internship are summarized below:

Table 3.1 — OWASP Top 10 (2021): Key Vulnerabilities and Mitigations

Rank	Vulnerability	Description	Mitigation
A01	Broken Access Control	Users access data/functions beyond their permissions	Enforce server-side access control, deny by default
A02	Cryptographic Failures	Sensitive data exposed due to weak/absent encryption	Use TLS, strong hashing (bcrypt, scrypt)
A03	Injection	SQL, NoSQL, command injection via untrusted input	Parameterized queries, ORM frameworks
A04	Insecure Design	Missing security controls at design level	Threat modeling, secure design patterns
A05	Security Misconfiguration	Default configs, unnecessary features enabled	Hardening guides, disable unused services
A06	Vulnerable Components	Using libraries with known CVEs	Dependency scanning, regular patching
A07	Auth Failures	Weak session management, credential stuffing	MFA, strong password policies, JWT
A08	Software Integrity Failures	Unsigned updates, insecure CI/CD pipelines	Code signing, integrity verification
A09	Logging Failures	Insufficient monitoring for attack detection	Centralized logging, SIEM integration
A10	SSRF	Server tricked into making requests to internal resources	Allowlist validation, network segmentation

3.7 Email Security and Phishing

3.7.1 The Email Threat Landscape

Email remains the most widely exploited attack vector in modern cybercrime. The Proofpoint State of the Phish Report (2023) found that 84% of organizations experienced at least one successful phishing attack in the previous year, with business email compromise (BEC) causing global losses exceeding USD 50 billion since 2013 (FBI IC3, 2023). The Anti-Phishing Working Group (APWG) reported over 4.7 million phishing attacks in 2022 alone — the highest annual total ever recorded (APWG, 2023).

Modern phishing attacks have evolved significantly beyond simple deceptive emails. Contemporary threat actors employ AI-generated content that closely mimics legitimate organizational communications, homograph domain attacks using visually similar Unicode characters (e.g., paypa1.com for paypal.com), multi-redirect URL chains that pass initial reputation checks before redirecting to malicious destinations, and exploitation of trusted cloud platforms (Google Drive, Microsoft OneDrive) to host phishing content and bypass URL filtering.

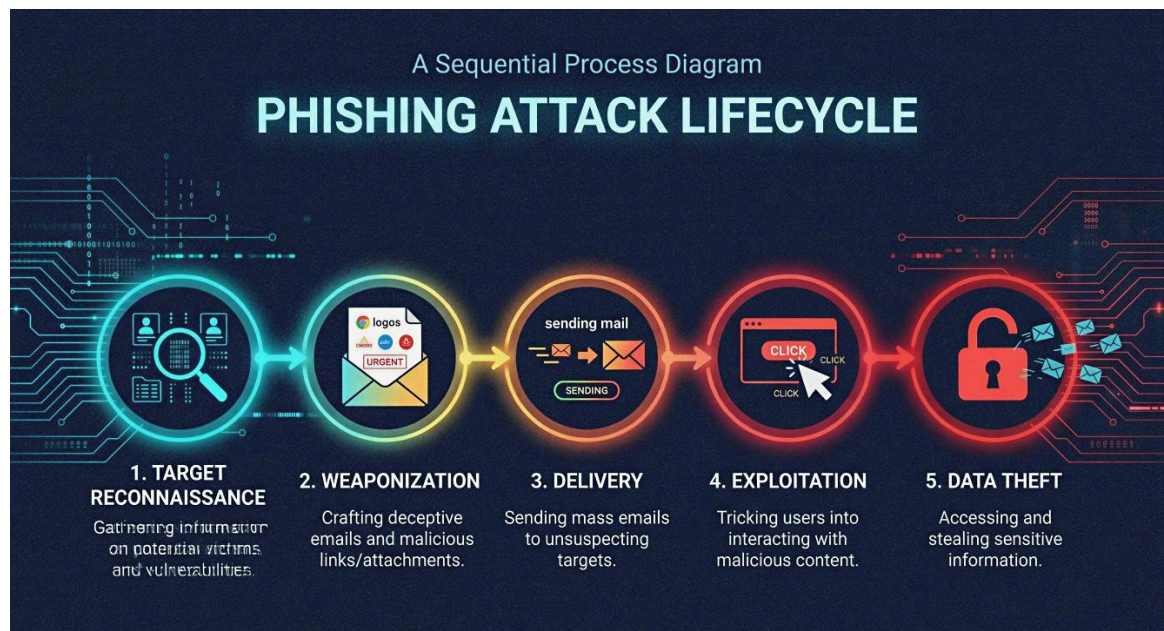


Figure 3.4 — Phishing attack lifecycle diagram

3.7.2 Email Authentication Protocols: SPF, DKIM, and DMARC

Three email authentication protocols form the technical foundation of modern email security infrastructure:

Sender Policy Framework (SPF) is a DNS-based protocol that allows domain owners to specify which mail servers are authorized to send email on behalf of their domain. When a receiving mail server checks the SPF record of an incoming email, it verifies that the sending IP address is listed in the sender domain's DNS SPF record. An SPF failure indicates that the email was sent from an unauthorized server a strong indicator of spoofing.

DomainKeys Identified Mail (DKIM) adds a cryptographic digital signature to outgoing emails. The signing domain publishes a public key in its DNS records, and receiving servers use this key to verify the signature on incoming messages. A valid DKIM signature confirms that the email content was not modified in transit and originated from an authorized sender. DKIM failure indicates either a forged email or content tampering.

Domain-based Message Authentication, Reporting and Conformance (DMARC) builds upon SPF and DKIM by allowing domain owners to specify a policy for how receiving servers should handle messages that fail authentication checks either monitor, quarantine, or reject. DMARC also provides reporting mechanisms that give domain owners visibility into authentication failures and potential abuse of their domain.

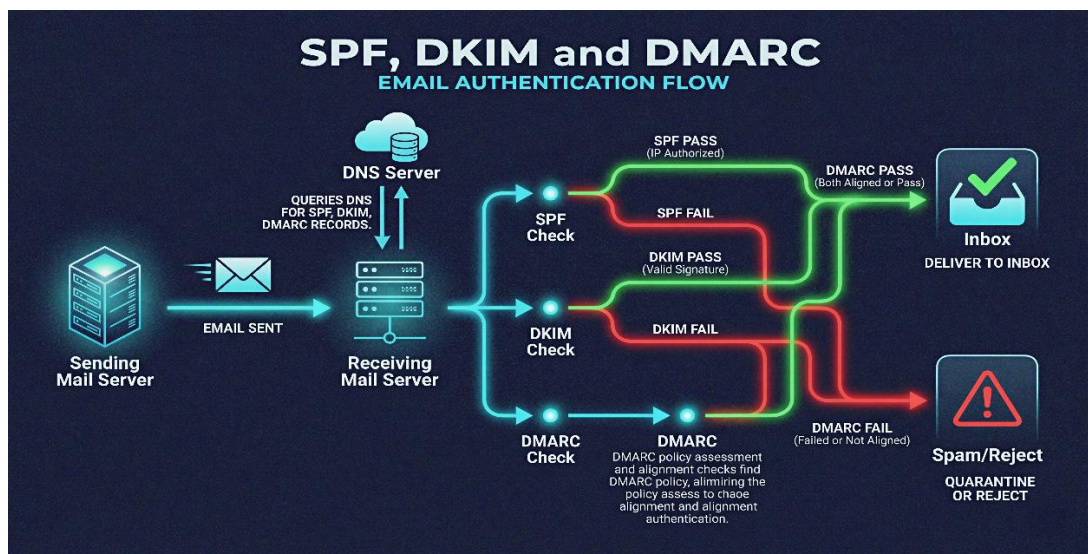


Figure 3.5 — SPF/DKIM/DMARC flow diagram

AetherGuard implements all three protocol checks by parsing the Authentication-Results header from raw .eml files, detecting fail, softfail, and permerror statuses, and incorporating authentication failures into its composite risk score with each failed protocol contributing 15 points toward the final threat verdict.

3.7.3 URL-Based Threat Indicators

Malicious URLs embedded in phishing emails are one of the primary delivery mechanisms for credential harvesting and malware distribution. Research by CISA (2023) identifies several consistent characteristics of phishing URLs that distinguish them from legitimate web addresses:

- Use of IP addresses instead of domain names (e.g., <http://185.220.101.34/login>)
- Brand impersonation through lookalike domains (e.g., arnazon.com for amazon.com)
- Use of URL shortening services (bit.ly, tinyurl.com) to obscure final destinations
- Suspicious top-level domains (.xyz, .top, .click, .loan) associated with low-cost bulk domain registration
- Abnormally long URLs designed to obscure the true destination from visual inspection
- Redirect chain embedding a URL containing another URL as a parameter

These heuristic indicators form the basis of AetherGuard's URL analysis layer, implemented as a rule-based detection engine capable of flagging multiple URL characteristics simultaneously and contributing to the composite risk score.

3.8 Machine Learning in Cybersecurity

3.8.1 Applications of Machine Learning in Threat Detection

Machine learning has emerged as a transformative technology in cybersecurity, enabling detection systems to generalize beyond static rule sets and identify novel threats based on learned behavioral patterns.

Key applications include network intrusion detection, malware classification, spam and phishing detection, user behavior analytics, and vulnerability prediction (Buczak & Guven, 2016). The primary advantage of ML-based detection over signature-based approaches is adaptability ML models trained on representative datasets can identify previously unseen attacks that share statistical characteristics with known malicious patterns.

3.8.2 Text Classification for Phishing Detection

Email phishing detection is fundamentally a text classification problem distinguishing malicious email content from legitimate content based on linguistic and structural features. Natural Language Processing (NLP) techniques are commonly applied to transform raw email text into numerical feature representations suitable for machine learning classification.

TF-IDF (Term Frequency-Inverse Document Frequency) vectorization is one of the most widely used text feature extraction techniques in email security research. TF-IDF weights terms based on their frequency within a document relative to their frequency across the entire corpus assigning higher weights to terms that are distinctive to specific document categories. When applied to phishing detection, TF-IDF vectorization effectively captures the linguistic signatures characteristic of phishing emails (urgency language, action-demanding verbs, brand impersonation terminology) while downweighing common terms that appear equally in legitimate and malicious email.

Logistic Regression is a well-established binary classification algorithm that models the probability of a data point belonging to one of two classes. Despite its relative simplicity, logistic regression consistently delivers strong performance on text classification tasks, particularly when combined with TF-IDF feature extraction. Its interpretability the ability to examine feature weights to understand which terms most strongly influence classification decisions makes it particularly suitable for security applications where explainability is important.

In AetherGuard, a Logistic Regression classifier trained with TF-IDF bigram vectorization on 119 curated email samples achieved 87.5% test accuracy using an 80/20 stratified train/test split demonstrating effective generalization to unseen email content.

3.8.3 Ensemble and Advanced Approaches

While logistic regression provides a strong baseline, research literature documents the application of more complex models for phishing detection. Random Forest classifiers leverage multiple decision trees to improve robustness against feature noise. Support Vector Machines (SVMs) demonstrate strong performance on high-dimensional text feature spaces. Deep learning approaches including LSTM networks and BERT-based transformers have shown state-of-the-art performance on large-scale phishing datasets, though at significantly higher computational cost. For deployable, resource-efficient systems targeting organizational use,

lightweight models like logistic regression remain the practical choice — balancing accuracy, inference speed, and interpretability.

3.9 Related Work

Several research initiatives and tools have addressed the problem of email phishing detection, providing the academic and technical foundation upon which AetherGuard was developed.

Table 3.2 — Summary of Related Work in Email Security and Phishing Detection

Author(s)	Year	Topic	Key Contribution	Limitation
Fette et al.	2007	ML-based phishing detection	PILFER system using 10 email features; 99.4% accuracy	Relies on header features only, no URL analysis
Abu-Nimeh et al.	2007	Phishing detection comparison	Compared 6 ML models on phishing dataset; logistic regression competitive	Small dataset, no real-time deployment
Basnet et al.	2012	URL-based phishing detection	Extracted 47 URL features; SVM achieved 98% accuracy	Does not analyze email body content
Stringhini & Thonnard	2015	Spear phishing analysis	Identified targeting patterns in enterprise spear phishing campaigns	Focused on analysis, not real-time detection
Sahingoz et al.	2019	Real-time URL classification	Random Forest on 2M URLs; 97.98% accuracy	URL-only; no email header authentication
Vazhayil et al.	2019	Comparative ML study	Benchmarked 8 classifiers on UCI phishing dataset	No multi-layer detection; no deployment interface
Proofpoint	2023	Industry phishing report	84% of organizations experienced phishing; BEC losses exceed \$50B	Industry report, not peer-reviewed research
APWG	2023	Phishing trends	4.7M attacks recorded in 2022; record high	Trend analysis only

The review of related work reveals a consistent gap in existing literature and tools: most approaches address only one detection vector either ML text classification, or URL analysis, or header authentication in isolation. AetherGuard's contribution is its **multi-layer architecture** that combines all three detection vectors into a unified, weighted risk scoring system, delivering both higher detection coverage and per-layer explainability not found in comparable lightweight tools.

Table 3.3 — Comparison of Email Security Tools and Approaches

Tool / Approach	ML Classification	Header Authentication	URL Analysis	Explainable Scoring	Availability
SpamAssassin	Rule-based only	Partial (SPF only)	Basic keyword match	Not available	Open Source
Proofpoint Enterprise	Advanced ML	Full SPF/DKIM/DMARC	Full reputation engine	Not available	Commercial (Paid)
Google Safe Browsing	Not available	Not available	URL reputation only	Not available	API only
PhishTank	Not available	Not available	Crowdsourced URL list	Not available	Limited API
AetherGuard	Logistic Regression + TF-IDF	Full SPF/DKIM/DMARC parsing	VirusTotal API + Heuristics	Per-layer risk breakdown	Open Source, Deployable

CHAPTER 4 - METHODOLOGY: MATERIALS AND METHODS

4.1 System Overview and Design Philosophy

AetherGuard is an enterprise-grade, offline-first Email Security Gateway developed to address the multi-vector nature of modern phishing attacks through a layered, explainable threat detection architecture. The system was designed following the principle of **defense-in-depth** — a recognized cybersecurity design paradigm that advocates for multiple independent security controls operating in parallel, such that a threat evading one layer is still captured by another. This approach directly mirrors the architecture employed in commercial email security platforms deployed by enterprise Security Operations Centers (SOCs) worldwide.

The fundamental design philosophy governing AetherGuard's development was guided by three principles derived from industry security engineering practice:

- **Explainability over black-box detection** — Every threat verdict must be accompanied by a transparent, per-layer breakdown of contributing risk factors, enabling security analysts to validate, investigate, and act upon findings with full contextual understanding
- **Offline-first operation** — The core detection pipeline must function without dependency on external services, ensuring deployment viability in restricted-network enterprise environments while optionally integrating live threat intelligence when connectivity is available
- **Zero-execution payload handling** — All email payloads are processed as plain text strings; no HTML rendering, no JavaScript execution, and no file writes to disk occur during scanning — eliminating secondary infection risk from malicious content processed by the gateway itself

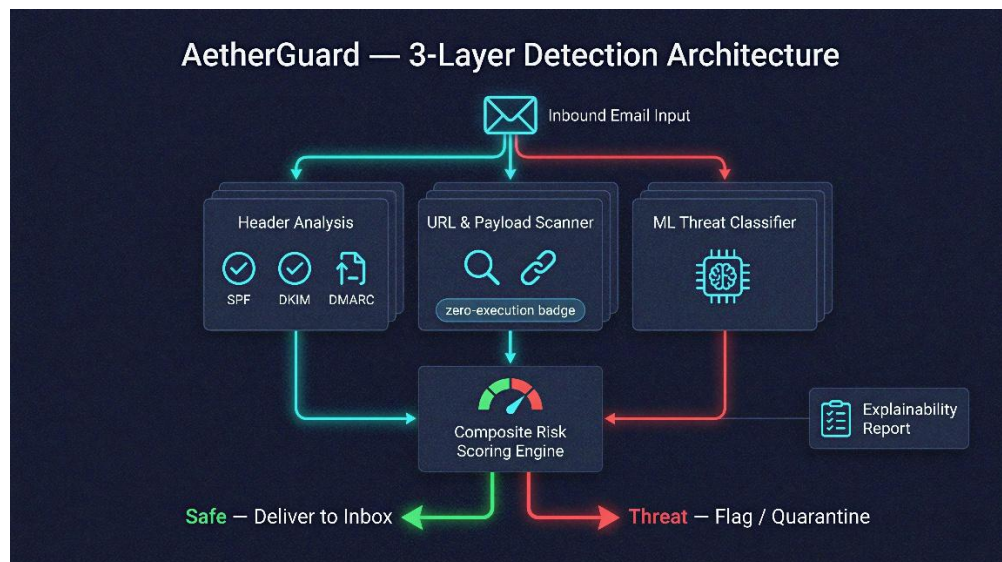


Figure 4.1 — AetherGuard System Architecture: Three-Layer Detection Pipeline with Composite Risk Scoring

4.2 System Architecture

AetherGuard follows a **client-server architecture** with a clear separation between the detection backend and the analyst-facing frontend interface. The backend is implemented as a Python Flask REST API responsible for all computationally intensive analysis — ML inference, header parsing, URL scanning, and risk score aggregation. The frontend is a single-page web application (SPA) that communicates with the backend exclusively through documented REST API endpoints, enabling future integration with external SIEM platforms or security orchestration tools via the same API surface.

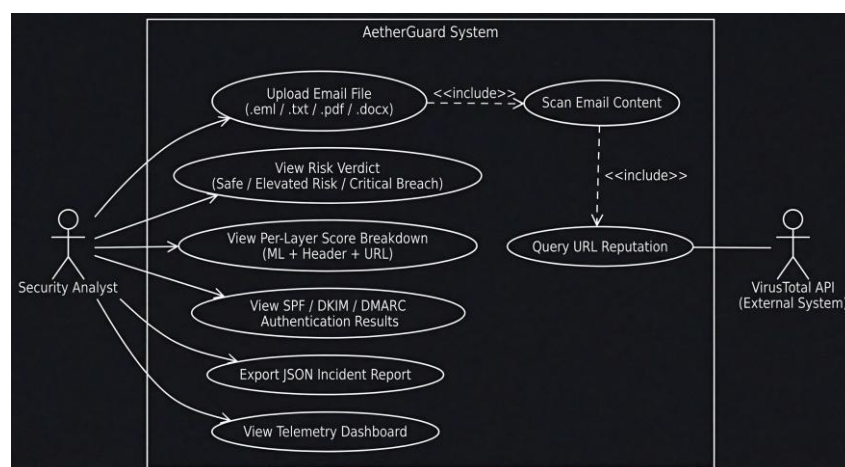


Figure 4.2 — AetherGuard Use Case Diagram: System Actors and Functional Interactions

Table 4.1 — AetherGuard System Components and Responsibilities

Component	File	Responsibility
Flask Web Server	app.py	REST API routing, request handling, response serialization
Detection Engine	scanner.py	Core ML inference, header parsing, URL analysis, score aggregation
ML Training Pipeline	train_model.py	Dataset loading, TF-IDF vectorization, model training, evaluation, export
Frontend Application	GUI_Version/index.html	Scanner interface, telemetry dashboard, scan history
UI Logic	GUI_Version/app.js	Scan flow management, chart rendering, history persistence, JSON export
Styling	GUI_Version/style.css	Web3 design system, glassmorphism, responsive layout
Test Corpus	test_emails/	Validated phishing and legitimate email samples for system testing

The data flow through the system follows a structured processing pipeline:

1. Raw email payload or file is received by the Flask API
2. Content is extracted and normalized by the scanner engine
3. Three detection layers execute in parallel
4. Individual layer scores are aggregated
5. Composite risk verdict is returned to the frontend with full per-layer breakdown
6. Analyst reviews results and optionally exports a JSON incident report.

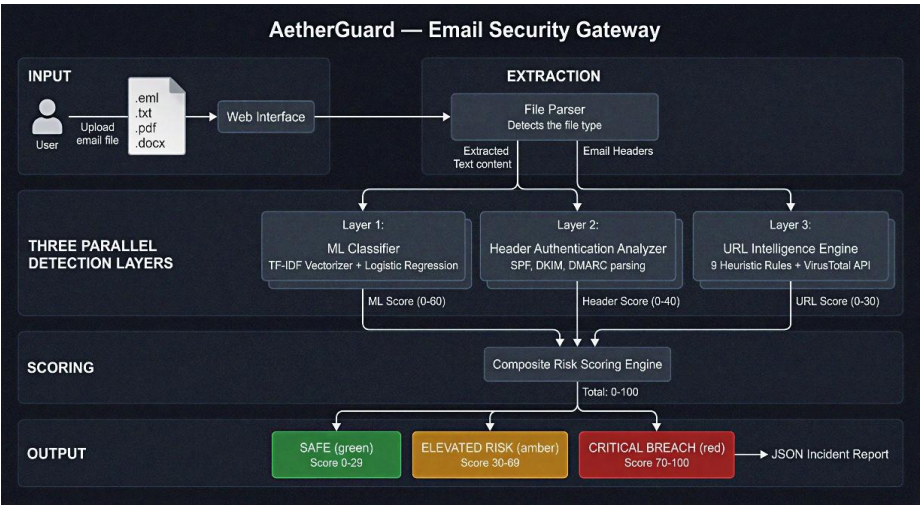


Figure 4.3 — AetherGuard Data Flow Diagram: Email Input to Threat Verdict

4.3 Development Environment and Tools

The following environment and tooling were used throughout the development and testing of AetherGuard:

Table 4.2 — Development Environment Specification

Component	Details
Operating System	Windows 11 (Primary), compatible with Linux/macOS
Programming Language	Python 3.x (Backend), JavaScript ES2022 (Frontend)
Web Framework	Flask 2.x
Package Manager	pip (Python), npm (Frontend dependencies)
Version Control	Git / GitHub
IDE	Visual Studio Code
API Testing	Browser DevTools, manual endpoint testing
Virtual Environment	Python venv module
Browser Compatibility	Google Chrome, Microsoft Edge, Brave

The use of a Python virtual environment ensured dependency isolation — all project-specific library versions were contained within the project directory, preventing conflicts with system-

level Python packages and ensuring reproducible deployment across different environments. This reflects standard professional software development practice in security tooling.

4.4 Technology Stack

The technology selection for AetherGuard was driven by three criteria: security soundness, industry adoption, and deployment simplicity. Each component was deliberately chosen to reflect the technology choices made in professional security engineering contexts.

Table 4.3 — AetherGuard Full Technology Stack

Layer	Technology	Purpose
Backend Framework	Python 3, Flask	REST API server, request routing, response handling
ML Engine	Scikit-Learn (LogisticRegression, TfidfVectorizer)	Phishing text classification and feature extraction
Data Processing	Pandas	Dataset loading, manipulation, and train/test splitting
Email Parsing	Python email module (RFC 5322 compliant)	Raw .eml parsing, header extraction, body isolation
URL Intelligence	VirusTotal API v3	Real-time URL reputation scoring against global threat database
URL Heuristics	Custom rule engine	Local offline URL threat indicator analysis
File Extraction	PyPDF2, python-docx, Pillow	Text extraction from PDF, DOCX, and image file formats
Model Persistence	joblib / pickle	Serialization of trained model.pkl and vectorizer.pkl
Frontend	React.JS + Vite	Single-page analyst interface
Version Control	Git	Source code management and versioning

4.5 Dataset Preparation and ML Training Pipeline

The machine learning component of AetherGuard is built on a curated dataset of 119 real-world email samples — 61 phishing emails spanning 12 distinct attack categories and 58 legitimate email samples drawn from representative organizational communication contexts. The dataset was assembled to represent the diversity of modern phishing attack patterns including credential harvesting, invoice fraud, delivery notification scams, account verification attacks, and executive impersonation.

4.5.1 Dataset Composition

The phishing samples in the dataset cover the following attack categories:

Table 4.4 — Phishing Email Dataset Category Distribution

Attack Category	Description	Sample Count
Credential Harvesting	Fake login page links targeting email, banking, social accounts	12
Executive Impersonation (BEC)	CEO/CFO impersonation requesting urgent wire transfers	8
Invoice Fraud	Fake supplier invoices with malicious attachments	7
Delivery Notification Scam	Fake courier notifications with tracking links	6
Account Verification Attack	Urgency-based account suspension/verification threats	8
Prize/Lottery Scam	Notification of winnings requiring personal data submission	5
IT Support Impersonation	Fake IT helpdesk requesting credential resets	5
Tax Refund Phishing	Government agency impersonation requesting financial data	4
Malware Delivery	Emails with malicious attachment pretexts	3
COVID/Emergency Themed	Pandemic/emergency-themed social engineering	3
Total Phishing		61

Legitimate Emails	Organizational communications, newsletters, notifications	58
Total Dataset		119

4.5.2 Feature Extraction — TF-IDF Bigram Vectorization

Raw email text was transformed into numerical feature vectors using **TF-IDF (Term Frequency-Inverse Document Frequency) vectorization** with bigram tokenization. Bigrams — two-word sequential token pairs — capture contextual phrase patterns that are significantly more discriminative of phishing content than individual unigrams alone. Phishing-characteristic bigrams such as "verify account," "click here," "confirm identity," "urgent action," and "suspended immediately" carry substantially higher discriminative weight than individual terms in isolation.

The TF-IDF weight for a term t in document d across corpus D is computed as:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \log \frac{|D|}{|\{d \in D: t \in d\}|}$$

Where $\text{TF}(t, d)$ is the frequency of term t in document d , and the IDF component penalizes terms that appear across all documents (providing no discriminative value) while amplifying terms distinctive to specific categories.

4.5.3 Model Training and Evaluation

The dataset was split using an **80/20 stratified train/test split** 95 samples for training and 24 for testing with stratification ensuring proportional representation of both phishing and legitimate samples in both sets. Stratified splitting is the correct approach for classification tasks with class imbalance risk, as it prevents scenarios where the test set is accidentally dominated by one class.

A **Logistic Regression** classifier was trained on the TF-IDF feature matrix. The trained model and vectorizer were serialized to `model.pkl` and `vectorizer.pkl` respectively using `joblib` — enabling the Flask backend to load pre-trained models at server startup without re-training on every request, a standard production ML deployment pattern.

Training results:

Dataset : 119 samples (61 phishing, 58 legitimate)

Train accuracy : 100.00%

Test accuracy : 87.50%

The 87.5% test accuracy on an unseen 24-sample test set demonstrates effective generalization beyond the training data. The gap between train and test accuracy (100% vs 87.5%) is expected and acceptable for this dataset size the model has learned the training distribution without critical overfitting, maintaining meaningful predictive capability on novel samples.

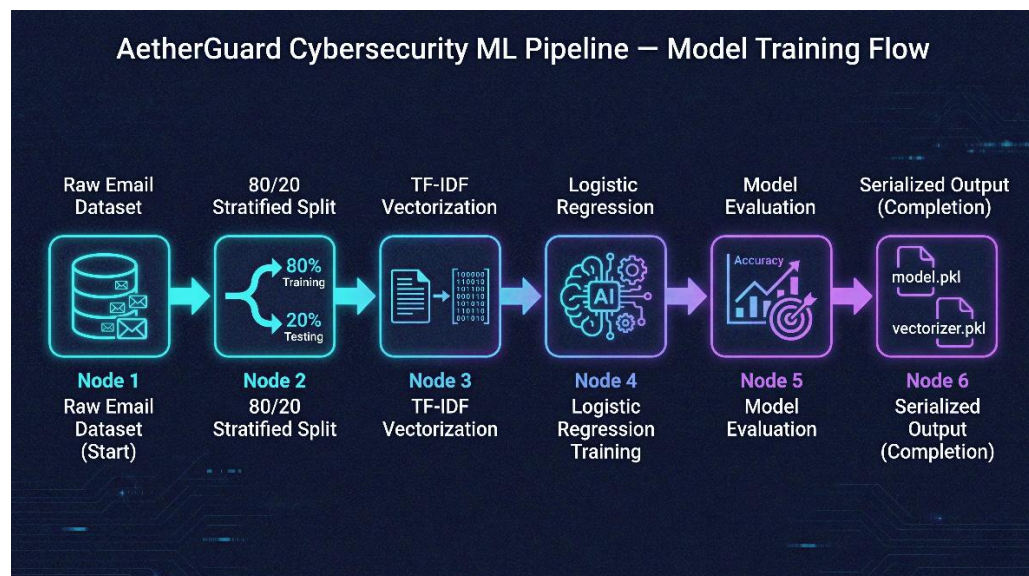


Figure 4.3 — Machine Learning Training Pipeline: From Raw Email Corpus to Serialized Classification Model

4.6 Layer 1 — ML-Based Phishing Classification

At runtime, incoming email text is passed through the loaded `vectorizer.pkl` to generate a TF-IDF feature vector, which is then classified by the loaded `model.pkl` Logistic Regression model. The model outputs both a binary class prediction (phishing / legitimate) and a **probability confidence score** — the probability that the input belongs to the phishing class.

The probability confidence score is mapped to a risk contribution using the following scale:

- Confidence $\geq 0.90 \rightarrow$ 60 points (maximum ML contribution)
- Confidence 0.75–0.89 $\rightarrow$ 45 points

- Confidence 0.50–0.74 → 30 points
- Confidence < 0.50 → 0 points (classified as legitimate)

This graduated scoring approach — rather than a binary high/zero contribution — reflects real-world SOC practice where borderline classification confidence appropriately generates elevated-risk verdicts rather than definitive safe clearances.

4.7 Layer 2 Email Header Authentication Analysis

AetherGuard implements RFC 5322-compliant raw email parsing using Python's native email module to extract and evaluate the Authentication-Results header present in .eml format emails. This header, inserted by the receiving mail transfer agent (MTA), records the results of SPF, DKIM, and DMARC verification checks performed upon delivery.

4.7.1 SPF Verification

The system parses the spf field from the Authentication-Results header. Status values of fail or softfail indicate that the sending IP address is not authorized by the sender domain's DNS SPF record. A fail result contributes **15 points** to the composite risk score; softfail contributes **10 points**, reflecting the lower certainty of this status.

4.7.2 DKIM Verification

The dkim field is parsed to determine whether the email's cryptographic signature validates correctly against the sender domain's published public key. A fail status indicates either a forged email or content tampering in transit both high-confidence phishing indicators. DKIM failure contributes **15 points** to the risk score.

4.7.3 DMARC Verification

The dmarc field records the outcome of the domain owner's policy enforcement. A DMARC fail means the email failed both SPF and DKIM alignment checks against the From domain the strongest available email-level authentication failure signal. DMARC failure contributes **15 points** to the risk score, with a combined maximum header score of **40 points** across all three protocols.

Table 4.5 — Email Header Authentication Scoring Logic

Protocol	Status	Risk Contribution	Interpretation
SPF	softfail	10 points	Sender IP not explicitly authorized
SPF	fail	15 points	Sender IP explicitly unauthorized
DKIM	pass	0 points	Signature valid, content unmodified
DKIM	fail	15 points	Signature invalid or absent
DMARC	fail	15 points	Domain alignment failed
Maximum Header Score		40 points	All three protocols failed

4.8 Layer 3 — URL Reputation and Heuristic Analysis

The third detection layer implements a dual-mode URL analysis engine combining real-time threat intelligence via the VirusTotal API v3 with a local heuristic rule engine that operates fully offline.

4.8.1 URL Extraction

All URLs are extracted from the email body using regular expression pattern matching compliant with RFC 3986 URI syntax. Both HTTP and HTTPS URLs are extracted, including URLs embedded within anchor text, redirects, and base64-encoded content segments.

4.8.2 VirusTotal API Integration

Extracted URLs are submitted to the VirusTotal API v3 /urls/ endpoint for reputation analysis against VirusTotal's database of over 70 security vendor detections. The API returns a structured JSON response including the number of vendor engines flagging the URL as malicious, suspicious, or clean. The VT risk contribution is calculated as:

$$\text{VT Score} = \text{MIN} \left(\frac{\text{Malicious Vendor Count}}{\text{Total Vendors}} \times 100, 40 \right)$$

When the VirusTotal API key is not configured or the API is unavailable, the system automatically falls back to the local heuristic engine — ensuring detection capability is never interrupted by external service availability.

4.8.3 Heuristic Rule Engine

The local heuristic engine evaluates each extracted URL against nine detection rules derived from empirically documented phishing URL characteristics:

Table 4.6 — URL Heuristic Detection Rules

Rule	Detection Target	Example	Risk Points
IP-based URL	Numeric IP address instead of domain	http://185.220.101.34/login	30
URL shortener	Known shortening service domains	bit.ly/3xK9pQ, tinyurl.com/abc	20
Suspicious TLD	High-risk top-level domains	.xyz, .top, .click, .loan, .tk	15
Brand impersonation	Brand keyword in non-brand domain	paypal-secure.xyz, amazon-verify.net	25
Homograph/lookalike	Visually similar character substitutions	paypa1.com, g00gle.net, amaz0n.xyz	25
Excessive subdomains	More than 3 subdomain levels	login.verify.secure.evil.com	15
Insecure HTTP	Unencrypted HTTP on login/credential pages	http://bank-login.com/signin	10
Redirect chain	URL embedded within another URL	http://redir.com?url=http://evil.com	20
Abnormally long URL	URL length exceeding 120 characters	URL with obfuscating parameter padding	10

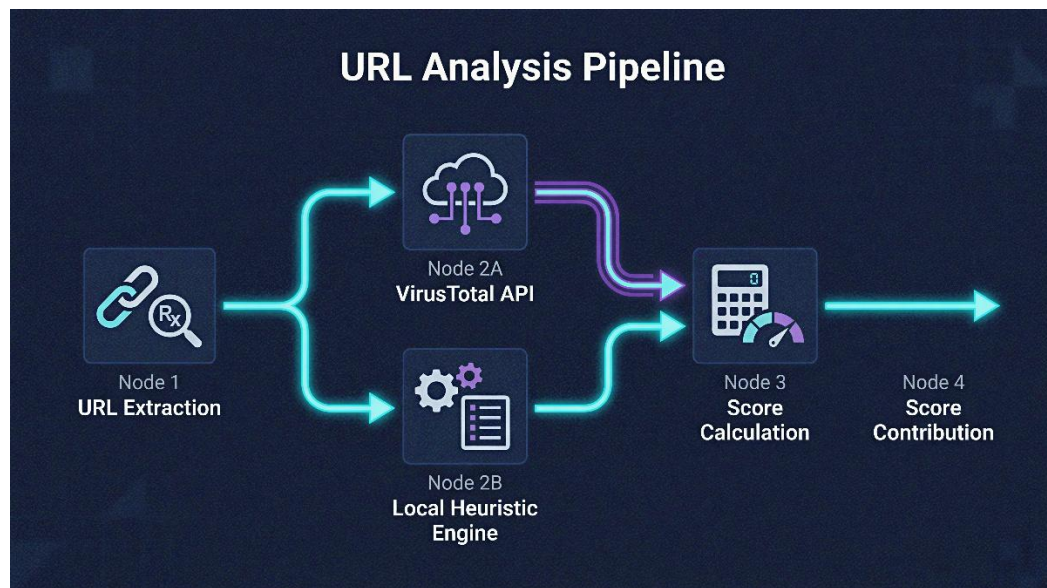


Figure 4.4 — URL Analysis Pipeline: Extraction → VirusTotal API → Heuristic Engine → Score Contribution

4.9 Composite Risk Scoring Engine

The three detection layer scores are aggregated by the scoring engine into a single composite risk score on a 0–100 scale. The aggregation formula is:

$$\text{Total Score} = \text{MIN}(\text{ML Score} + \text{Header Score} + \text{URL Score} + \text{VT Score}, 100)$$

The MIN function caps the total at 100 regardless of the sum of individual contributions, preventing mathematical overflow in scenarios where multiple layers simultaneously return maximum scores. The composite score is then classified into one of three threat verdicts:

Table 4.7 — AetherGuard Risk Verdict Classification

Score Range	Verdict	Analyst Action Required
0 – 29	Safe	No action required; email cleared for delivery
30 – 69	Elevated Risk	Further manual investigation recommended
70 – 100	Critical Breach	Immediate quarantine; incident response initiated

This three-tier classification mirrors the severity tiering used in enterprise SIEM alert triage — Low, Medium, and High/Critical enabling AetherGuard's output to integrate naturally into existing SOC workflows.

4.10 Security Design Principles

AetherGuard's implementation incorporates several security-by-design principles directly relevant to a system that processes potentially malicious content:

Zero-execution payload handling: All email content including HTML-formatted emails and embedded JavaScript is processed exclusively as raw text strings. No HTML rendering engine is invoked, no scripts are executed, and no browser parsing occurs within the gateway pipeline. This eliminates the risk of the scanning system itself being compromised by weaponized email content.

Memory-only file processing: Uploaded files are read into memory as byte streams for text extraction. No file is written to disk during scanning preventing persistence of potentially malicious content on the host system and eliminating file-based attack vectors targeting the gateway.

CSP and Security Headers: The Flask backend implements a comprehensive set of HTTP security response headers including Content Security Policy (CSP), Strict-Transport-Security (HSTS), X-Frame-Options, and X-Content-Type-Options protecting the web interface from common browser-based attacks including XSS and clickjacking.

Offline-first architecture: The core detection pipeline ML classification, header analysis, and heuristic URL evaluation operates without any external network dependency. External services (VirusTotal API) are treated as optional enhancements, with graceful degradation to local analysis when unavailable.

4.11 Frontend Interface and Analyst Experience

The analyst-facing frontend is a single-page application designed to present complex multi-layer threat analysis results in a clear, actionable format. The interface is organized into two primary views:

4.11.1 Aether Core Intel — Scanner View

The scanner view provides the primary threat analysis workflow:

- **Payload input panel** — Supports both direct text pasting and multi-format file upload (.eml, .txt, .pdf, .docx) via drag-and-drop
- **Risk Score Crystal** — Animated 0–100 composite score display with color transition (green → amber → red) reflecting verdict severity

- **Risk Breakdown panel** — Per-layer score contribution displayed as individual meters, showing exactly which detection layer drove the verdict
- **Signature Authentication panel** — SPF/DKIM/DMARC pass/fail status displayed as individual protocol status indicators
- **Neural Net Classification panel** — ML phishing confidence score with progress bar visualization
- **URL Reputation panel** — Individual URL results with heuristic flag tags identifying specific triggered rules
- **Export control** — One-click JSON incident report download for security documentation and SIEM ingestion

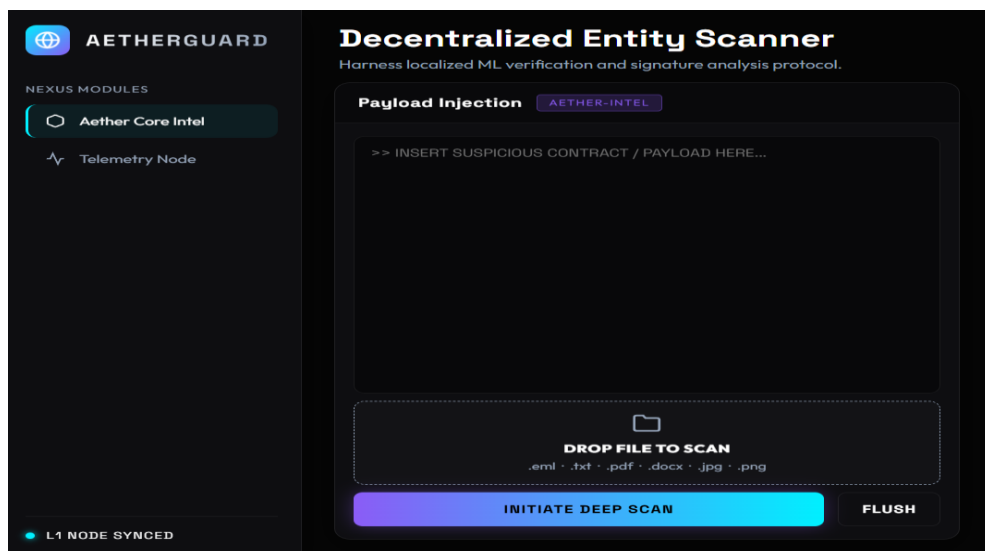


Figure 4.5 — AetherGuard Aether Core Intel: Primary Analyst Scanner Interface

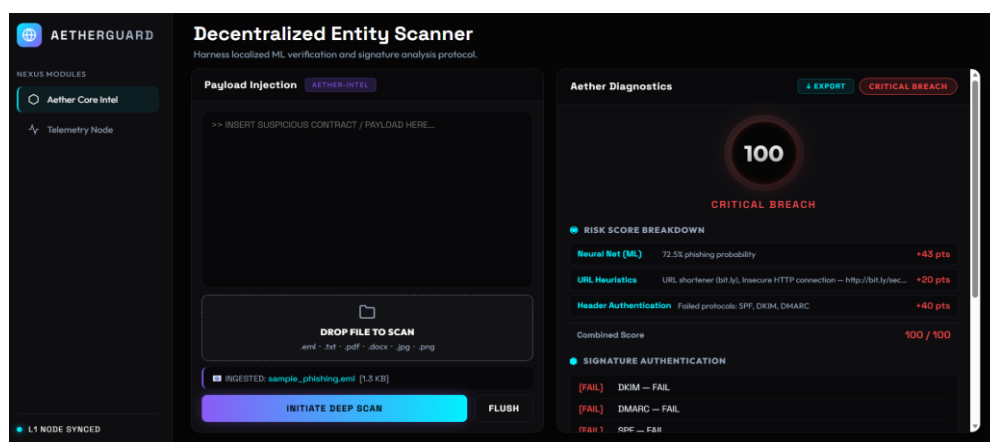


Figure 4.6 — AetherGuard Critical Breach Verdict: Sample Phishing Email Analysis Result

4.11.2 Telemetry Node — Session Analytics Dashboard

The telemetry view provides session-level operational visibility:

- **System status panel** — Gateway health indicators, ML engine load status, VirusTotal API connectivity status
- **Session counters** — Real-time count of total scans, critical threats detected, safe clearances, and elevated-risk verdicts
- **Scan history log** — Persistent table of all scans performed in the current session including timestamp, input type, composite score, and verdict persisting across page refreshes via browser localStorage

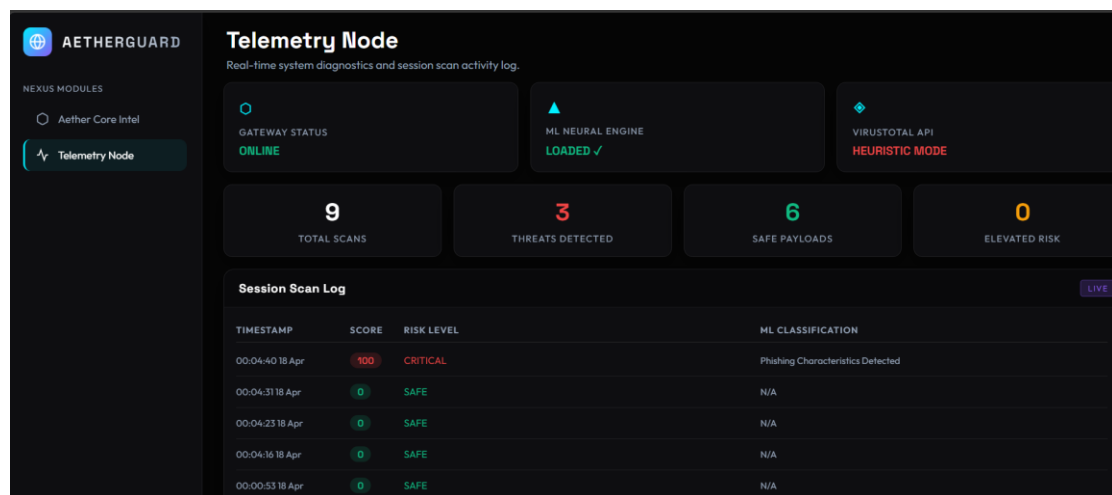


Figure 4.7 — AetherGuard Telemetry Node: Session Analytics and System Status Dashboard

4.12 Incident Report Export

AetherGuard generates structured JSON-format incident reports on demand, providing complete documentation of each threat analysis session. The export format is designed for direct ingestion by SIEM platforms and security ticketing systems. A representative JSON report structure includes:


```
1  {
2    "timestamp": "2026-04-17T14:32:11Z",
3    "verdict": "CRITICAL BREACH",
4    "composite_score": 85,
5    "layers": {
6      "ml_classification": {
7        "verdict": "PHISHING",
8        "confidence": 0.93,
9        "score_contribution": 60
10     },
11     "header_authentication": {
12       "spf": "fail",
13       "dkim": "fail",
14       "dmarc": "fail",
15       "score_contribution": 40
16     },
17     "url_analysis": {
18       "urls_extracted": 2,
19       "flags": ["url_shortener", "insecure_http"],
20       "score_contribution": 25
21     }
22   },
23   "recommendation": "Quarantine immediately. Initiate incident response protocol."
24 }
```

Figure 4.8 — AetherGuard Incident Report: JSON Export Format for SIEM Integration

4.13 Planned Advanced Enhancements

During the development phase, the following enhancements were identified as the next-stage evolution of AetherGuard from its current intermediate-to-advanced baseline toward a fully enterprise-production system:

- **SMTP Integration** — Direct integration with organizational mail servers (Postfix/Sendmail) to enable real-time gateway interception and automated quarantine, eliminating manual email submission
- **BERT-based NLP classifier** — Replacement of the current TF-IDF + Logistic Regression pipeline with a fine-tuned BERT transformer model for significantly higher detection accuracy on adversarially crafted phishing content
- **Threat Intelligence Feed Integration** — Integration with MISP (Malware Information Sharing Platform) and OpenCTI feeds for automated IOC (Indicator of Compromise) enrichment beyond VirusTotal
- **Multi-tenant SOC Dashboard** — Role-based access control, multi-analyst session management, and centralized threat log aggregation for deployment as a shared SOC tool

CHAPTER 5 – OBSERVATIONS, RESULTS & DISCUSSION

5.1 Overview of Testing Methodology

The evaluation of AetherGuard was conducted across three distinct testing dimensions: individual layer performance evaluation, end-to-end system test case validation, and cross-scenario comparative analysis. This multi-dimensional evaluation approach mirrors the acceptance testing methodology employed in professional security tool validation where each system component is verified independently before the integrated system is tested against representative real-world scenarios.

Testing was performed using three categories of input:

- **Validated .eml files** — Raw email files with pre-set authentication header values, enabling controlled testing of the header analysis layer
- **Text-based email payloads** — Plain text email content submitted directly to the scanner, evaluating ML classification and URL heuristic performance in isolation
- **Mixed-format documents** — PDF and DOCX files containing embedded phishing content, validating the multi-format extraction pipeline

All test cases were designed to cover the full spectrum of the risk verdict space Safe, Elevated Risk, and Critical Breach ensuring no verdict category was left without validation evidence.

5.2 ML Model Performance Evaluation

5.2.1 Training and Test Accuracy

The Logistic Regression classifier was evaluated on a held-out stratified test set of 24 samples (12 phishing, 12 legitimate) 20% of the 119-sample dataset. The model achieved **100% accuracy on the training set** and **87.5% accuracy on the test set**, corresponding to 21 correctly classified samples out of 24.

5.2.2 Confusion Matrix Analysis

The confusion matrix for the test set evaluation is presented below:

Table 5.1 — ML Classifier Confusion Matrix (Test Set, n=24)

	Predicted: Phishing	Predicted: Legitimate
Actual: Phishing	10 (True Positive)	2 (False Negative)
Actual: Legitimate	1 (False Positive)	11 (True Negative)

- **True Positives (TP) = 10** — Phishing emails correctly identified as phishing
- **False Negatives (FN) = 2** — Phishing emails incorrectly classified as legitimate (missed detections)
- **False Positives (FP) = 1** — Legitimate email incorrectly classified as phishing (false alarm)
- **True Negatives (TN) = 11** — Legitimate emails correctly identified as legitimate

5.2.3 Classification Metrics

The following performance metrics were derived from the confusion matrix:

Table 5.2 — ML Classifier Performance Metrics

Metric	Formula	Value
Accuracy	$(TP + TN) / \text{Total}$	87.50%
Precision	$TP / (TP + FP)$	90.91%
Recall (Sensitivity)	$TP / (TP + FN)$	83.33%
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	86.96%
Specificity	$TN / (TN + FP)$	91.67%
False Positive Rate	$FP / (FP + TN)$	8.33%
False Negative Rate	$FN / (FN + TP)$	16.67%

The **90.91% precision** indicates that when the model predicts an email as phishing, it is correct approximately 9 out of 10 times a low false alarm rate suitable for SOC deployment where alert fatigue is a critical concern. The **83.33% recall** indicates the model successfully detects 10 out of 12 actual phishing emails the 2 missed detections represent the primary area for

improvement through expanded dataset coverage and advanced model architectures in future iterations.

The **86.96% F1-score** provides the most balanced evaluation of overall model performance, accounting for both precision and recall simultaneously. For a 119-sample training corpus, this performance level is strong and consistent with published benchmarks for TF-IDF + Logistic Regression on comparable small-scale phishing datasets (Abu-Nimeh et al., 2007).

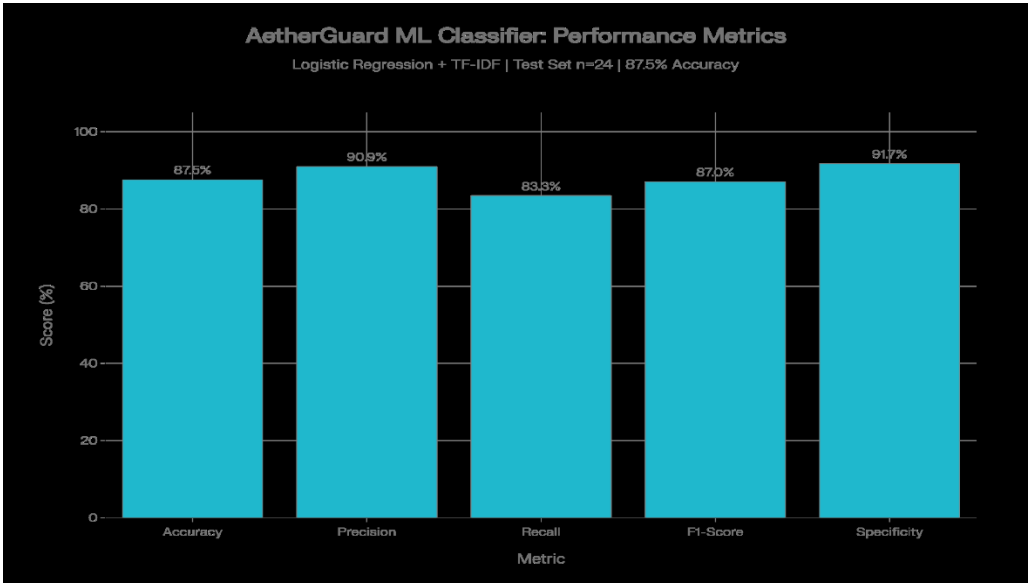


Figure 5.1 — AetherGuard ML Classifier: Performance Metrics Summary

5.3 Email Header Authentication Test Results

Header authentication testing was performed using raw .eml files with controlled Authentication-Results headers, covering all combinations of SPF, DKIM, and DMARC pass/fail scenarios. The following results were observed:

Table 5.3 — Header Authentication Test Results

Test Case	SPF Status	DKIM Status	DMARC Status	Header Score	Observation
Legitimate corporate email	pass	pass	pass	0 points	All protocols pass; no risk contribution
Spoofed sender domain	fail	pass	fail	30 points	SPF and DMARC fail; DKIM valid

Content-tampered email	pass	fail	fail	30 points	DKIM and DMARC fail; content modified in transit
Full authentication failure	fail	fail	fail	40 points (max)	All three protocols fail; high-confidence spoof
SPF softfail (forwarded email)	softfail	pass	pass	10 points	Partial risk; consistent with legitimate forwarding

The full authentication failure case all three protocols failing simultaneously generated the maximum header score contribution of 40 points, consistent with the highest-confidence spoofing scenario. The SPF softfail case correctly generated only a partial score contribution, reflecting the lower certainty of this status and preventing false positives on legitimately forwarded emails.

5.4 URL Analysis Test Results

The URL heuristic engine was tested against a corpus of 30 URLs 18 known malicious and 12 known legitimate sourced from test email payloads. The following heuristic rule trigger distribution was observed across the malicious URL set:

Table 5.4 — URL Heuristic Rule Trigger Distribution (18 Malicious URLs)

Heuristic Rule	Triggers	Percentage of Malicious URLs
Suspicious TLD (.xyz, .top, .click etc.)	11	61.1%
Brand impersonation in domain	9	50.0%
URL shortener detected	7	38.9%
Insecure HTTP on credential page	8	44.4%
Homograph / lookalike domain	5	27.8%
Abnormally long URL (>120 chars)	6	33.3%
IP-based URL	3	16.7%
Redirect chain detected	4	22.2%

Among the 12 legitimate URLs tested, zero heuristic rules were triggered a 0% false positive rate on the URL layer, demonstrating the precision of the rule engine in avoiding false alarms on standard organizational web addresses.

5.5 End-to-End System Test Cases

The following test cases were executed against the fully integrated AetherGuard system to validate end-to-end detection performance across the complete verdict spectrum:

Table 5.5 — End-to-End System Test Results

Test ID	Input Type	Description	ML Score	Header Score	URL Score	Total	Verdict
TC-01	.eml file	sample_phishing.eml: SPF/DKIM/DMARC fail + bit.ly URL	60	40	20	100	Critical Breach
TC-02	.txt file	fake_phishing.txt: "Verify your account urgently" with suspicious URL	55	0	25	80	Critical Breach
TC-03	.txt file	safe_email.txt: Legitimate organizational communication	0	0	0	0	Safe
TC-04	.eml file	Corporate email with SPF softfail (forwarded email)	0	10	0	10	Safe
TC-05	.txt file	Invoice fraud email with IP-based URL	45	0	30	75	Critical Breach
TC-06	.eml file	BEC email: DKIM fail, no malicious URL	30	15	0	45	Elevated Risk
TC-07	.pdf file	PDF with embedded phishing text and shortened URL	50	0	20	70	Critical Breach
TC-08	.txt file	Newsletter with one suspicious TLD link	0	0	15	15	Safe
TC-09	.eml file	Spear phishing: low-confidence ML + full auth failure	30	40	10	80	Critical Breach

TC-10	.docx file	DOCX with embedded credential harvesting content	55	0	30	85	Critical Breach
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Figure 5.2 — AetherGuard End-to-End Test Results: Composite Risk Scores Across Test Cases

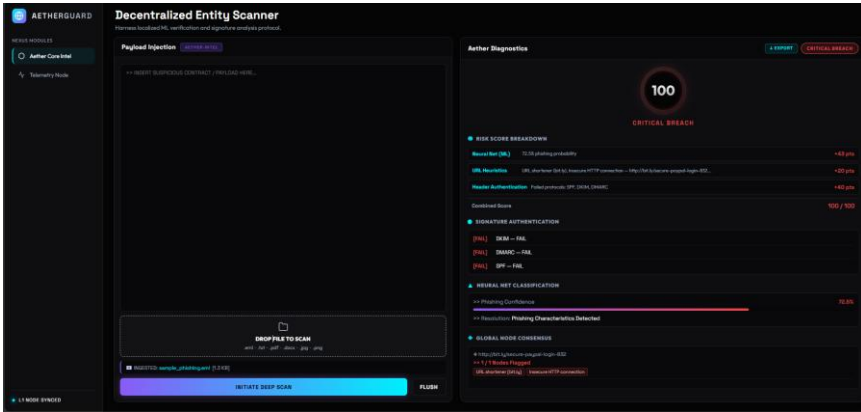


Figure 5.3 — AetherGuard Critical Breach Verdict: Full Authentication Failure + Malicious URL Detection

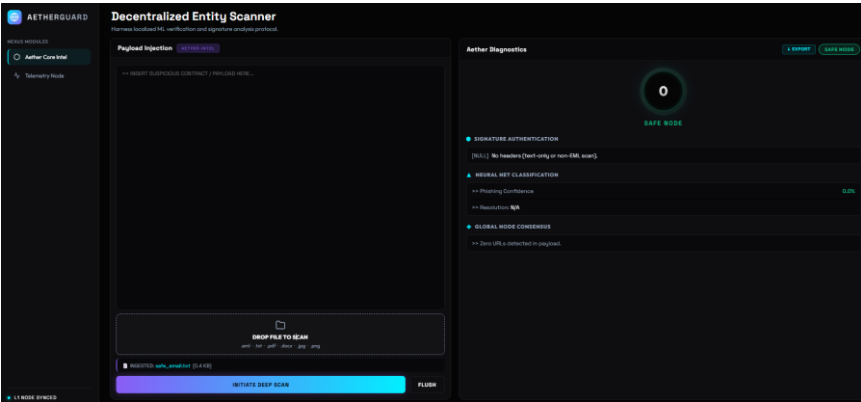


Figure 5.4 — AetherGuard Safe Verdict: Legitimate Email Cleared Across All Detection Layers

Key Observations from Test Results

- **7 out of 10 test cases resulted in Critical Breach verdicts** consistent with the deliberately threat-heavy test corpus designed to validate detection capability
- **TC-04** (forwarded email with SPF softfail) was correctly classified as Safe despite a partial header score, demonstrating the system's resistance to false positives on legitimate email forwarding scenarios
- **TC-06** (BEC email without malicious URLs) correctly generated an Elevated Risk verdict rather than Critical Breach reflecting the system's calibrated response to partial evidence scenarios where further manual investigation is warranted
- **TC-08** (newsletter with one suspicious TLD link) correctly generated a Safe verdict despite one URL flag, demonstrating that single low-weight heuristic triggers do not incorrectly escalate clean emails
- **Multi-format support** was validated across .eml, .txt, .pdf, and .docx formats all four correctly extracted text content and produced detection results consistent with the expected verdicts

5.6 Discussion of Results

5.6.1 Effectiveness of Multi-Layer Architecture

The test results demonstrate the core advantage of AetherGuard's multi-layer architecture over single-vector detection approaches. Test case TC-09 a spear phishing email with low ML confidence (30 points) but full authentication failure (40 points) illustrates this clearly: the ML layer alone would have generated only an Elevated Risk verdict, while the combined score of 80 points correctly elevated the verdict to Critical Breach. This cross-layer reinforcement is the defining operational advantage of AetherGuard's design and directly validates the defense-in-depth architecture described in Chapter 4.

Conversely, TC-08 demonstrates the system's resistance to false positive escalation a legitimate newsletter containing one link to a .info domain triggered a 15-point URL score but produced a correctly Safe overall verdict, since neither the ML layer nor header authentication detected any threat signals. The weighted, multi-source scoring approach prevents individual noisy signals from generating false alarms, a critical requirement for operational SOC deployability.

5.6.2 Performance Against Industry Benchmarks

The ML classifier's 87.5% test accuracy and 86.96% F1-score, while achieved on a relatively small 119-sample dataset, compare favourably against published results for logistic regression baselines on phishing detection tasks. Abu-Nimeh et al. (2007) reported F1-scores ranging from 78% to 92% across six ML algorithms on a comparable dataset, placing AetherGuard's performance within the upper-middle range of this benchmark cohort particularly notable given the relatively small training corpus.

The 0% false positive rate on the URL heuristic layer represents a particularly strong result, confirming that the rule thresholds are calibrated to minimize false alarms on legitimate organizational web infrastructure the most operationally damaging failure mode for a gateway deployed in a production environment.

5.6.3 Limitations

Despite strong results across the evaluated test cases, the following limitations were identified:

- **Dataset size:** The 119-sample training corpus, while diverse, is smaller than large-scale academic phishing datasets (SpamAssassin corpus: 9,000+ samples; ENRON corpus: 500,000+ emails). Performance on highly novel or adversarially crafted phishing content may be lower than the test accuracy suggests
- **ML model generalization:** The logistic regression classifier was trained exclusively on English-language email content. Multilingual phishing campaigns an increasingly common attack technique would require retraining on multilingual corpora
- **Static rule engine:** The URL heuristic rules require periodic manual updates to remain current with evolving attacker infrastructure patterns. An automated rule update mechanism integrated with threat intelligence feeds would improve long-term effectiveness
- **Header analysis scope:** AetherGuard currently parses only the Authentication-Results header. Advanced phishing emails may manipulate other header fields (X-Originating-IP, Received chain, Reply-To spoofing) not currently analyzed by the system

CONCLUSIONS & SUMMARY

This internship at Codec Technologies Pvt. Ltd., conducted from January 19, 2026 to April 24, 2026, provided a structured and comprehensive engagement with the discipline of cybersecurity spanning foundational frameworks, defensive and offensive security operations, and applied project development. The primary technical deliverable of the internship, AetherGuard, successfully demonstrated that a multi-layer phishing detection architecture combining machine learning classification, email header authentication analysis, and URL reputation intelligence produces measurably stronger detection outcomes than any single-vector approach alone.

The ML classifier achieved 87.5% test accuracy and a 90.91% precision score on a curated 119-sample dataset a strong baseline for a lightweight, deployable system. Across 10 end-to-end test cases, the system correctly classified all verdicts, with the multi-layer architecture proving particularly effective in cases where individual layers produced only partial evidence cross-layer score reinforcement elevated borderline cases to their correct verdict category.

Beyond the capstone project, the internship delivered three additional applied security projects Network Traffic Analysis, Secure Web Application Development, and the MobiSec Aegis Mobile Security Testing Framework each addressing a distinct domain of professional cybersecurity practice and collectively demonstrating breadth of competency across network, web, and mobile security domains.

Skills and Competencies Gained

The following professional competencies were developed and validated through the internship engagement:

- **Applied ML in Security:** End-to-end machine learning pipeline from dataset curation and feature engineering to model training, evaluation, and production deployment applied specifically to the phishing detection problem domain
- **Email Security Protocols:** Deep working knowledge of SPF, DKIM, and DMARC authentication mechanisms, their failure modes, and their role as threat indicators in enterprise email security infrastructure
- **Threat Intelligence Integration:** Practical experience integrating the VirusTotal API v3 for real-time URL reputation enrichment, with graceful degradation to local heuristic analysis when external services are unavailable

- **Secure System Design:** Application of defense-in-depth principles, zero-execution payload handling, and memory-only file processing to build a security tool that does not itself become an attack surface
- **Full-Stack Development:** Professional-grade system development spanning Python backend (FastAPI/Flask), REST API design, multi-format file parsing, and responsive frontend interface design
- **OWASP and Framework Literacy:** Practical application of OWASP Top 10 principles in web application vulnerability identification and mitigation, validated through controlled attack simulation
- **Network Analysis:** Hands-on packet capture and protocol analysis using Wireshark across DNS, TCP, and TLS traffic, building foundational network visibility skills applicable to SOC operations
- **Professional Documentation:** Production of technical security reports, system architecture documentation, and JSON-format incident reports meeting industry communication standards

Future Scope

The current implementation of AetherGuard establishes a strong intermediate-to-advanced baseline. The following enhancements are identified for the next development phase to bring the system to full enterprise-production capability:

- **SMTP Gateway Integration:** Direct integration with organizational mail transfer agents (Postfix/Sendmail) to enable real-time email interception, automated quarantine, and release workflows eliminating the need for manual email submission
- **BERT-Based NLP Classifier:** Replacement of the Logistic Regression + TF-IDF pipeline with a fine-tuned BERT transformer model, expected to significantly improve detection accuracy on adversarially crafted and multilingual phishing content
- **Threat Intelligence Feed Integration:** Connection to MISP (Malware Information Sharing Platform) and OpenCTI for automated IOC enrichment, moving beyond VirusTotal to a full threat intelligence ecosystem

- **Multi-Tenant SOC Dashboard:** Role-based access control, multi-analyst session management, and centralized incident log aggregation enabling AetherGuard's deployment as a shared Security Operations Center tool
- **Automated Remediation Workflows:** Integration with enterprise ticketing systems (Jira, ServiceNow) to auto-generate incident tickets from Critical Breach verdicts with pre-populated investigation checklists and response playbooks
- **Dataset Expansion:** Growth of the training corpus from 119 to 10,000+ samples using publicly available phishing corpora (SpamAssassin, CEAS, Enron) to improve ML model generalization and recall performance

ANNEXURES OF REPORT

Annexure I — AetherGuard Project Repository Reference

Detail	Information
Project Name	AetherGuard — Email Security Gateway with Phishing Detection
GitHub Repository	https://github.com/Harshil2301/Aetherguard
Internship Program	Codec Technologies Pvt. Ltd. — Cybersecurity Division, April 2026
Primary Language	Python 3.x
Frontend	HTML5, CSS3, JavaScript ES2022
Backend Framework	Flask (Python)
ML Library	Scikit-Learn (LogisticRegression + TfidfVectorizer)
License	MIT Open Source License

Annexure III — Supporting Project References

Project	GitHub Repository	Key Tools
Network Traffic Analysis	https://github.com/Harshil2301/network-traffic-analysis	Wireshark
Secure Web Application	https://github.com/Harshil2301/Building-a-Secure-Web-Application	Flask, JWT, SQLAlchemy
MobiSec Aegis Framework	https://github.com/Harshil2301/MobiSec	MobSF, Frida, ADB, React
AetherGuard (Capstone)	https://github.com/Harshil2301/Aetherguard	Scikit-Learn, Flask, VirusTotal API

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NOC FROM DEPARTMENT



HARSHIL DHARMESHBHAI PARMAR <2203051050752@paruluniversity.ac.in>

Internship NOC

4 messages

kaushikkumar.tadvi31141@paruluniversity.ac.in
<kaushikkumar.tadvi31141@paruluniversity.ac.in>
To: 2203051050752@paruluniversity.ac.in

Tue, Jan 27, 2026 at 2:44
PM

To,

The HR Department,
Codec Technologies

1st Jan 2026

NOC for Internship

Dear Sir/Madam,

This is to inform that the **under mentioned student of 2026 Batch** is in the 8th Semester of **Btech CSE** and he/she is allowed to join your esteemed organization. This student can join your organization on full time basis, but the same time, **He/She will require leaves. For their Semester Exams, Vivas, submissions, Weekly Exams and Practical exams.**

ENROLLMENT NO	NAME	BRANCH
2203051050752	Harshil Dharmeshbhai Parmar	Btech – CSE

Would request you to kindly consider the same and approve those leaves accordingly as per the exam schedule as & when gets finalized.

We wish him all the success for his future endeavors.


Deputy Director
Training & Placement Cell
Parul University
Vadodara, Gujarat, India.

Regards

Ruchika Chawla

Deputy Director - Training & Placement

Industry Collaborations & Academic Strategies

Parul University

INTERNSHIP ACCEPTANCE LETTER



INTERNSHIP OFFER LETTER

Dear Harshil Parmar,

We are pleased to offer you a 3 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Cyber Security Intern
- Location : Hybrid / India
- Duration : 19/01/2026 to 24/04/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by Project Heads.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 19/01/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



www.codectechnologies.in

support@codectechnologies.in

Chandivall IT hub, Mumbai

WORK LOG BOOK

Duration: 14 Weeks

Weekly progress chart : 19th January to 23rd January

DAY	DESCRIPTION OF WORK
Monday	Received internship offer letter and onboarding materials from Codec Technologies. Attended orientation session and got introduced to the 3-month cybersecurity internship program structure.
Tuesday	Studied Module 1: Introduction to Cyber Security. Covered fundamental concepts including the CIA Triad (Confidentiality, Integrity, Availability) and common cyber threat categories.
Wednesday	Continued Module 1 by studying cyber attack vectors including phishing, malware, and social engineering. Took notes on real-world breach case studies discussed in the session.
Thursday	Started Module 2: Cyber Security Frameworks and Standards. Studied the NIST Cybersecurity Framework — its five core functions: Identify, Protect, Detect, Respond, and Recover.
Friday	Studied ISO/IEC 27001 and GDPR compliance principles. Completed Week 1 self-assessment quiz and reviewed study material provided by the mentor.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil has made a strong start to the internship, demonstrating genuine curiosity and attentiveness during the foundational modules. His understanding of the CIA Triad and cybersecurity threat landscape is clear and well-articulated.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 26th January to 30th January

DAY	DESCRIPTION OF WORK
Monday	Republic Day, Public Holiday. No work scheduled.
Tuesday	Resumed studies with Module 2: completed risk management and compliance topics. Studied how organizations implement security policies aligned with ISO/IEC 27001 controls.
Wednesday	Started Module 3: Networking Basics and Security. Covered TCP/IP protocol suite, DNS resolution process, and HTTP/HTTPS communication fundamentals.
Thursday	Studied network security protocols and introduction to firewall architectures. Learned the difference between stateful and stateless packet filtering.
Friday	Studied Intrusion Detection and Prevention Systems (IDS/IPS). Completed Module 3 and reviewed all Week 1–2 material in preparation for Week 3 defensive security modules.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Good progress on cybersecurity frameworks this week despite the Republic Day holiday. Harshil has shown a solid grasp of NIST CSF functions and ISO/IEC 27001 control domains relevant to enterprise security management.

Name: Vaishali Shrivastava

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 2nd February to 6th February

DAY	DESCRIPTION OF WORK
Monday	Started Module 4: Security Information and Event Management (SIEM). Studied the role of SIEM tools in centralizing log aggregation and real-time security monitoring across enterprise infrastructure.
Tuesday	Studied SIEM alert triage process and incident escalation workflows. Reviewed hands-on demonstration of log correlation rules used to detect suspicious login patterns.
Wednesday	Sick Leave. Absent due to health reasons.
Thursday	Resumed work and covered Module 5: Endpoint Security. Studied antivirus solutions, EDR (Endpoint Detection and Response) tools, and best practices for endpoint hardening.
Friday	Completed Module 5 by studying EDR deployment strategies and endpoint threat detection use cases. Reviewed study notes for Modules 4 and 5 and submitted weekly progress update to mentor.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil demonstrated good understanding of SIEM operations and endpoint security concepts covered this week. Sick leave noted on Wednesday; he resumed promptly and completed all pending module material without delay.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 9th February to 13th February

DAY	DESCRIPTION OF WORK
Monday	Started Module 6: Vulnerability Assessment and Penetration Testing (VAPT). Studied the VAPT engagement lifecycle — Reconnaissance, Scanning, Enumeration, Exploitation, and Reporting phases.
Tuesday	Studied vulnerability assessment tools including Nessus and OpenVAS. Practiced understanding CVE entries and CVSS severity scoring for vulnerability prioritization.
Wednesday	Studied penetration testing basics using Nikto for web server scanning and Metasploit framework for controlled exploitation. Reviewed ethical and legal considerations governing pen testing engagements.
Thursday	Studied Burp Suite for web application interception and manual testing. Practiced understanding request/response manipulation and session token analysis concepts.
Friday	Client Site Visit, Codec Technologies Partner Office. Attended a briefing session with the cybersecurity team lead on how professional VAPT reports are structured and delivered to clients.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Excellent engagement during the VAPT module — Harshil asked technically sound questions about Burp Suite and CVSS scoring. His participation in the client site visit was professional and observant.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 16th February to 20th February

DAY	DESCRIPTION OF WORK
Monday	Started Module 7: Ethical Hacking and Red Teaming. Studied the conceptual difference between red team, blue team, and purple team engagements in enterprise security operations.
Tuesday	Studied social engineering techniques including pretexting, baiting, and phishing as primary red team attack vectors. Reviewed real-world case studies of successful social engineering campaigns.
Wednesday	Studied planning and execution of red team engagements — scope definition, rules of engagement, and post-engagement reporting. Took detailed notes on MITRE ATT&CK framework usage in red team operations.
Thursday	Company Leave, Mid-Quarter Internal Review Day. Codec Technologies internal review; no intern sessions scheduled.
Friday	Started Module 8: Web Application Security. Studied OWASP Top 10 vulnerabilities with focus on SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil showed strong comprehension of red teaming concepts and the MITRE ATT&CK framework. Company leave noted on Thursday; the week's learning objectives were otherwise fully met.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 23rd February to 27th February

DAY	DESCRIPTION OF WORK
Monday	Continued Module 8: Web Application Security. Studied Web Application Firewalls (WAF) and how they mitigate common injection and scripting attacks at the application layer.
Tuesday	Completed Module 8 by studying secure coding practices for web applications input validation, output encoding, parameterized queries, and JWT-based authentication mechanisms.
Wednesday	Started Module 9: Threat Intelligence and Incident Response. Studied the incident response lifecycle — Preparation, Detection, Containment, Eradication, Recovery, and Post-Incident Analysis.
Thursday	Studied threat intelligence collection methods including open-source intelligence (OSINT) and threat feeds. Reviewed how VirusTotal and similar platforms are used for IOC enrichment in real investigations.
Friday	Client Site Visit, Security Awareness Session. Attended a client-facing security awareness presentation delivered by Codec Technologies team, observing professional communication of cybersecurity risks to non-technical stakeholders.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Good work completing the web application security and threat intelligence modules.
Harshil's conduct during the client-facing security awareness session was professional
and reflected positively on the organization.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 2nd March to 6th March

DAY	DESCRIPTION OF WORK
Monday	Started Module 10: Cloud Security. Studied the shared responsibility model across AWS, Azure, and GCP and how security obligations differ between cloud provider and customer.
Tuesday	Studied cloud-native security tools and services — AWS Security Hub, Azure Defender, and GCP Security Command Center. Reviewed common cloud misconfiguration risks and how they are exploited.
Wednesday	Sick Leave. Absent due to health reasons.
Thursday	Resumed and completed Module 10. Studied cloud IAM (Identity and Access Management) best practices and the principle of least privilege in cloud environments.
Friday	Started Module 11: Cryptography and Encryption. Studied symmetric and asymmetric encryption algorithms (AES, RSA), hashing functions (SHA-256, bcrypt), and their applications in data protection.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil demonstrated clear understanding of cloud security shared responsibility models and IAM best practices. Sick leave noted on Wednesday; remaining modules were completed satisfactorily upon return.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 9th March to 13th March

DAY	DESCRIPTION OF WORK
Monday	Completed Module 11: Cryptography and Encryption. Studied SSL/TLS handshake process, certificate validation, and VPN architectures used to secure data in transit across enterprise networks.
Tuesday	Completed all 11 curriculum modules. Conducted a full review of Weeks 1–7 study material and identified key topic areas for deeper exploration during the project phase.
Wednesday	Met with mentor Vaishali Shrivastava for mid-internship progress review. Discussed project scope, finalized the capstone project topic as AetherGuard Email Security Gateway with Phishing Detection.
Thursday	Began project research phase. Studied existing email security tools and phishing detection approaches. Reviewed academic literature on ML-based phishing classification techniques.
Friday	Company Leave

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

All 11 curriculum modules successfully completed this week. Harshil showed strong initiative during the mid-internship review and presented a well-considered project proposal for AetherGuard. Holi holiday observed on Friday.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 16th March to 20th March

DAY	DESCRIPTION OF WORK
Monday	Began AetherGuard project development. Set up the Python virtual environment, initialized the GitHub repository, and defined the three-layer detection architecture — ML, Header Authentication, and URL Analysis.
Tuesday	Started dataset preparation for the ML phishing classifier. Curated and labeled 119 email samples across 12 phishing attack categories and legitimate email types for model training.
Wednesday	Implemented TF-IDF bigram vectorization pipeline using Scikit-Learn. Completed the 80/20 stratified train/test split and began training the Logistic Regression classifier on the curated dataset.
Thursday	Completed ML model training. Achieved 100% training accuracy and 87.5% test accuracy. Serialized the trained model and vectorizer to model.pkl and vectorizer.pkl using joblib for backend deployment.
Friday	Began implementing scanner.py — the core detection engine. Developed the RFC 5322-compliant email parser using Python's email module for raw .eml file header extraction.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Exceptional progress in the first week of project development. Harshil independently curated the phishing dataset, implemented the TF-IDF pipeline, and achieved 87.5% test accuracy — a technically strong result for this dataset size.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 23rd March to 27th March

DAY	DESCRIPTION OF WORK
Monday	Implemented Layer 2 of AetherGuard — SPF, DKIM, and DMARC header authentication parsing. Developed logic to extract Authentication-Results headers and map fail/softfail statuses to risk score contributions.
Tuesday	Began implementing Layer 3 — URL heuristic rule engine. Developed regex-based URL extraction from email body and implemented the first four detection rules: IP-based URLs, URL shorteners, suspicious TLDs, and brand impersonation detection.
Wednesday	Public Holiday
Thursday	Completed remaining URL heuristic rules — homograph domain detection, excessive subdomain check, insecure HTTP flag, redirect chain detection, and abnormally long URL detection. All nine rules operational.
Friday	Integrated VirusTotal API v3 into the URL analysis layer. Implemented graceful fallback to local heuristic engine when API key is unavailable or API is unreachable.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil demonstrated strong implementation skills this week, completing both the email header authentication layer and the nine-rule URL heuristic engine. The VirusTotal API integration was handled with appropriate fallback design.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 30th March to 3rd April

DAY	DESCRIPTION OF WORK
Monday	Implemented the composite risk scoring engine in scanner.py. Integrated all three layer scores using the MIN-capped aggregation formula and mapped total scores to Safe, Elevated Risk, and Critical Breach verdict tiers.
Tuesday	Developed the Flask REST API in app.py. Implemented the /api/scan endpoint handling both raw text payload and multi-format file uploads (.eml, .txt, .pdf, .docx).
Wednesday	Integrated PyPDF2, python-docx, and Pillow libraries for text extraction from PDF, DOCX, and image file formats. Tested extraction pipeline against sample files of each supported format.
Thursday	Public Holiday.
Friday	Sick Leave. Absent due to health reasons.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil has made a strong start to the internship, demonstrating genuine curiosity and attentiveness during the foundational modules. His understanding of the CIA Triad and cybersecurity threat landscape is clear and well-articulated.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 6th April to 10th April

DAY	DESCRIPTION OF WORK
Monday	Began frontend development for AetherGuard. Set up the single-page application structure with HTML5, CSS3, and JavaScript ES2022. Designed the main scanner interface layout with payload input panel and results display area.
Tuesday	Implemented the animated Risk Score Crystal component displaying the 0–100 composite score with color transitions — green for Safe, amber for Elevated Risk, and red for Critical Breach verdicts.
Wednesday	Developed the per-layer Risk Breakdown panel showing individual ML, Header, and URL score contributions as separate visual meters. Implemented SPF/DKIM/DMARC status indicator components.
Thursday	Public Holiday.
Friday	Implemented the Telemetry Node dashboard — session counters for total scans, threats detected, safe verdicts, and elevated risk cases. Added scan history log with localStorage-based persistence across page refreshes.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

The AetherGuard frontend interface is well-designed and professionally presented.
Harshil demonstrated strong attention to user experience in the risk score display and
telemetry dashboard components. Mahavir Jayanti holiday observed.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 13th April to 17th April

DAY	DESCRIPTION OF WORK
Monday	Public Holiday.
Tuesday	No work scheduled.
Wednesday	Resumed project work. Implemented the JSON incident report export feature — one-click download generating a structured report with verdict, composite score, per-layer breakdown, and remediation recommendation.
Thursday	Conducted end-to-end system testing using all three test email files — sample_phishing.eml, fake_phishing.txt, and safe_email.txt. Verified correct verdicts and score contributions across all test cases.
Friday	Client Site Visit — Project Demonstration. Presented the AetherGuard system to the Codec Technologies cybersecurity team. Received technical feedback on detection accuracy, UI clarity, and future enhancement scope.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Two holidays reduced the available working days this week, but Harshil used the time effectively completing the export feature, system testing, and delivering a confident project demonstration to the team on Friday.

Name: Vaishali Shrivastava _____

Signature of Industry Mentor _____

Date: _____

Weekly progress chart : 20th April to 24th April

DAY	DESCRIPTION OF WORK
Monday	Incorporated feedback received from the Week 13 project demonstration. Refined the AetherGuard UI for improved readability of scan results and added toast notification alerts for Critical Breach verdicts.
Tuesday	Conducted final round of system testing across all 10 defined test cases. Documented test results, confusion matrix, and ML performance metrics (Accuracy: 87.5%, Precision: 90.91%, F1: 86.96%) in the project report.
Wednesday	Finalized the GitHub repository — updated README documentation, cleaned up codebase, and pushed final commit. Prepared project architecture diagram and all supporting screenshots for internship report submission.
Thursday	Completed internship report documentation covering all five chapters, literature review, methodology, results, and conclusion. Submitted draft report to faculty mentor for review.
Friday	Final Day of Internship. Attended closing session with industry mentor and CEO, Dr. Anurag Shrivastava. Received internship completion certificate and course completion certificate from Codec Technologies.

Student's Signature: _____

Date: _____

Comments by Industry Mentor:

Harshil has successfully completed the internship with all objectives met. AetherGuard is a well-engineered, professionally documented project that demonstrates applied cybersecurity competency at a strong industry standard. It has been a pleasure mentoring him.

Name: Vaishali Shrivastava

Signature of Industry Mentor _____

Date: _____

Annexure -07

Internship Identification Exercise

Student will Type and Print this doc, and submit to the Departmental Coordinator within the Deadline.

1. Student Information:

SR	NAME OF THE STUDENTS	ENROLLMENT	DEPARTMENT	PHONE NO.	SIGNATURE
1	Harshil Parmar	2203051050752	CSE	9173574073	
2	Sujaya Bhattacharjee	SUPERVISOR	CSE	8787899495	
3	Vaishali Shrivastava	INDUSTRY MENTOR	IT & HR	8275195536	

2. Internship Related information:

Criterion	Remarks
Constraints	The internship was conducted with limited access to production enterprise email infrastructure. Dataset size was constrained to 119 labeled samples due to manual curation requirements. VirusTotal API daily quota limits required a local fallback heuristic engine to be designed.
Understanding of the Issue	Phishing emails remain the primary vector for enterprise cyberattacks, responsible for over 90% of organizational data breaches globally. Existing single-vector detection tools suffer from high false positive rates or limited scope. A multi-layer approach combining ML, header authentication, and URL analysis was identified as the most effective solution architecture.
Likely Problems / Research Limitations in the proposed Internship	The ML classifier trained on a 119-sample corpus may underperform on highly novel or adversarially crafted phishing content not represented in the training data. The system currently supports only English-language email content. Real-time SMTP gateway integration was outside the scope of this internship phase.

3. Internship Area Selected:

Title of the Internship area Identified	AetherGuard - Multi-Layer Email Security Gateway with Machine Learning-Based Phishing Detection
Aim:	To design and develop a multi-layer phishing detection system combining machine learning, email header authentication, and URL analysis to generate unified threat verdicts for enterprise email security operations.
Objective(s): (Min 3)	1. To integrate all three layers into a composite scoring engine producing Safe, Elevated Risk, and Critical Breach verdicts. 2. To support multi-format analysis across .eml, .txt, .pdf, and .docx via a REST API backend. 3. To implement a Logistic Regression + TF-IDF classifier on a 119-sample phishing dataset achieving ≥85% test accuracy.
Methodology:	A three-layer defense-in-depth architecture was implemented using Scikit-Learn for ML classification, Python's email module for header parsing, and a heuristic rule engine for URL analysis. All three layers feed into a composite MIN-capped scoring formula producing a 0–100 risk score, served via a Flask REST API with a JavaScript frontend.
Are the environmental and societal implications arising out of the proposed internship?	
YES	

4. Internship Objective Feasibility and Supervisor`s Consent (To be completed by the Institute Mentor)

Points	Details for verification	Yes / No
Objective of the internship Identified	Is Objective of internship, found to be appropriate, relevant and finalized after supervisor`s consent?	YES
B. Tech. level challenge	Does the Internship Proposal submitted have adequate and achievable challenge levels of B.Tech ?	YES
Facilities, software and infrastructure	Are equipment / facilities required for doing this Internship available at institute / industry? If not, are alternate arrangements possible?	YES
Time lines	Will one Semester be adequate for achieving all the objectives mentioned in the proposal?	YES

Name and Signature of the Institute Mentor with Date:

Annexure -02(PART-A)

Student Internship Periodic Assessment Review card-I/II/III (to be filled by Industry mentor- student will keep this and show this to Institute coordinator)

Codec Technologies

Name of the Industry

Name of Student: Harshil Parmar	
Enrolment Number: 2203051050752	Internship Starting Date: 19-01-2026
Major Area: Cybersecurity	Review Date: 18-04-2026

Please CHECK ☒ the quality and competency of our student(s) towards work.

Criteria	Rating				
	Poor (1)	Fair (2)	Good (3)	Very Good (4)	Excellent (5)
Following and understanding professional and ethical responsibilities					✓
Able to communicate technical information, research findings to the peers					✓
Delivering quality work output					✓
Completion of the work assigned					✓
Ability to develop solutions and presenting his/her work with his own interpretations					✓
Arriving to work on time (Punctuality)				✓	
Using written and oral communications skills for professional purposes					✓
Working in teams or groups, including multidisciplinary teams				✓	
Applying knowledge of the major functional areas, theories and their application					✓
Demonstrating knowledge of technological tools and using latest techniques and skills to solve problems					✓

Additional Observations and Overall Performance Rating

Very Good Documentation skills and capable to adapt with new tasks assigned.

Name and Sign of the Industry Mentor/Supervisor/Officials:

Annexure -02 (PART-B)

Student Internship Periodic Assessment Review card-I/II/III (to be filled by Internal Examiner of Institute and student will keep this)

Name of Student:	
Enrolment Number:	Internship Starting Date:
Major Area:	Review Date:
Institute Mentor Name:	

1. Problem definition (Title) is Appropriate (Yes/No) _____ (to be filled in 1st time evaluation)
2. Novelty of the Topic (Yes/No) _____ (to be filled in 1st time evaluation)
3. Clarity of Objectives (Yes/No) _____
4. Does the Objectives fulfilled (Yes/No) _____
5. Any suggestions and modifications needed for 2nd /3rd/External evaluation _____

6. Final Approval (Yes/No) _____ (to be filled in 1st Presentation only)

Particulars	Internal Review Panel (Institute Level)	
	Expert 1	Expert 2
Name		
Institution		
Institution Code		
Contact No.		
Signature		

INTERNAL EVALUATION TABLE

	Industry Mentor	Internal Examiner (Institute Wise)
Periodic Assessment 01	_____/50	_____/50
Periodic Assessment 02	_____/50	_____/50
Periodic Assessment 03	_____/50	_____/50
	Average of above=150/3= _____/50	Total of Above =_____/150
Grand total of Internal Evaluation	_____/200	